

Review article

Comprehensive Analysis on Tomato: Disease Detection, Recommendations, and Advanced Technological Approaches in AI**Gangadhar Shankarappa* and Pramod Tumakuru Channabasavanna***Department of Computer Science and Engineering, Siddaganga Institute of Technology, Tumkur, India*

Received: 1 July 2025, Revised: 29 January 2026, Accepted: 12 February 2026, Published: 14 May 2026

Abstract

Tomato plants are vulnerable to a wide range of diseases that affect their growth and crop yield, resulting in significant economic losses for farmers. This paper provides a comprehensive analysis of tomato plant life cycle and environmental conditions required for optimum crop growth. It explores tomato varieties along with their properties, detailed information on diseases across all the growth stages, including causes, symptoms, and control measures. The paper also provides an extensive evaluation of different artificial intelligence methods for tomato plant disease detection and prediction. It is observed that the detection accuracy of AI models ranges from 92% to 99%. Additionally, the paper explores advanced technologies like drones/UAV, IoT, cloud and edge computing, and blockchain for tomato cultivation management. The advantages and challenges w.r.t each technology are analyzed. The insights and solutions discussed in this paper can help farmers and researchers effectively manage tomato plant diseases and better understand supporting technologies that can enhance productivity and enable disease prediction for sustainable tomato cultivation.

Keywords: tomato disease detection; artificial intelligence; machine learning; deep learning; IoT; cloud computing; block chain technology

1. Introduction

Tomatoes are a high-value crop that provides substantial income for small and marginal farmers in India. The relatively short growing season allows for numerous harvests in a year, ensuring a stable income stream. India is one of the major producers of tomatoes, with significant export potential to adjacent countries and further, contributing to foreign exchange earnings. The inclusion of tomatoes in crop management cycles plays a significant role in soil health improvement, pest management, and agricultural productivity (Sharma et al., 2023). Most of the processed food industries rely on tomatoes, and they are used in the production of a wide range of products including ketchup, puree, and sauces across the world. Tomato prices are subject to sudden change, which can have a

*Corresponding author: E-mail: gangadhar.smg24@gmail.com

<https://doi.org/10.55003/cast.2026.268213>

Copyright © 2024 by King Mongkut's Institute of Technology Ladkrabang, Thailand. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

significant impact on farmers and consumers. Attempts to control tomato prices can only occur if there is proper storage, transportation, and other market conditions for economic balance.

Tomato production also plays an important role in pushing innovation and improved methods in agricultural technology. Protected cultivation (greenhouses and polyhouses), drip irrigation, and integrated pest management involving genetically produced pest resistant varieties have improved tomato production and resource utilization. The highlighted innovations are not only beneficial for increasing yield, but also for minimizing agriculture's overall contribution to environmental degradation (Sotelo-Cardona et al., 2021). Government incentives and subsidies (Asia Farming, 2025) in the tomato industry have led to the advanced practices and have contributed towards the general modernization of the sector. Such a force towards sustainable and efficient methods of agricultural production conforms to India's large policy objectives of sustainable and efficient farming and food security system that will enable future generations to continue growing tomatoes in a way that is lucrative for them.

In the 2023-24 season, the area of India under tomato cultivation was approximately 809,000 hectares (8090 km²), which is part of the broader horticultural expansion perceived in the latest years (Ministry of Agriculture and Farmers Welfare, Government of India, 2024). For the 2022-23 accounting year, India produced around 20.4 million metric tons of tomatoes. The production estimate for 2023-24 was slightly higher at approximately 20.8 million metric tons, indicating a stable and slightly upward trend in tomato production (Press Information Bureau, 2024). Tomato farming is profitable for farmers. For example, the average cost of cultivating tomatoes on one acre is around ₹30,150, while the revenue from marketing the crop can be significantly higher, providing a good return on investment. Exact profit margins may differ depending on yield, market prices, and local conditions. The agricultural sector contributed 18.3% to the Gross Value Added (GVA) of India as of 2022-23 (Saxena et al., 2023). While specific data on tomatoes alone isn't isolated in national GDP figures, horticulture (which includes tomatoes) is a major component of this sector. It is estimated that the global tomato market would grow at a compound annual growth rate (CAGR) of 7.2% during the period 2023-2032 and will reach a value of about USD 328.9 Bn by 2032 from USD 167 Bn in 2022 (MarketResearch.biz, 2025) as shown in Figure 1.

Significant volatility was brought to light by the abrupt surge in tomato prices from ₹20 per kg in June 2022 to ₹100 per kg in July 2023. This price increase, together with a negative inflation rate for tomatoes, is part of a phenomenon known as Tomato-nomics. About 50% of all tomatoes produced in India are grown in the states of Andhra Pradesh, Madhya Pradesh, Karnataka, Odisha, and Gujarat. Climatic conditions like extreme weather, attacks by different types of insects, and viruses like Tomato Mosaic Virus (ToMV) and Cucumber Mosaic Virus (CMV) have inflated the prices and decreased the profit of the farmers and they have been forced to switch to other crops. Solutions to this include improving supply chains, expanding processing capacity, improving direct sales and supporting controlled environment farming. Figure 2 illustrates tomato prices and inflation in India from 2018 to 2023 (Drishiti IAS, 2023).

Tomato farming requires advanced technologies to face the ever rising demand for high quality produce without compromising sustainability and efficiency. Technologies like IoT based sensors and drone monitoring offer precision agriculture that help real time monitoring of soil conditions, plant health and environmental factors. Machine learning and AI techniques are used to improve disease detection and forecast yield to minimize losses and maximize resource utilization. Innovations such as improving productivity, managing climate related risks, reducing water and fertilizer use, and encouraging more eco-friendly practice are crucial in modern tomato cultivation (Girish et al., 2018).

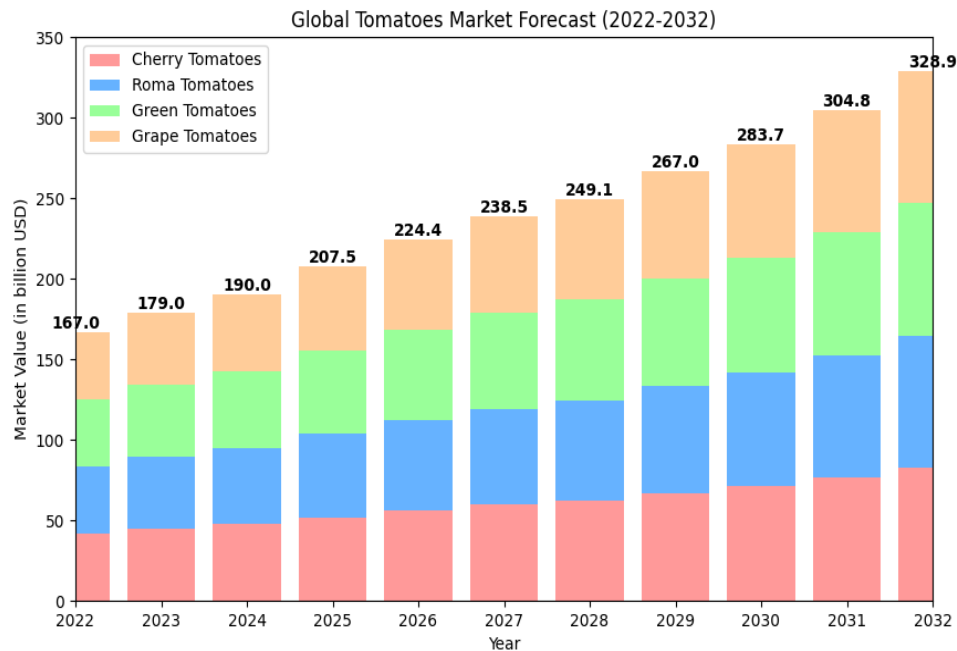


Figure 1. Global tomatoes market forecast (2022-2032) (MarketResearch.biz, 2025)

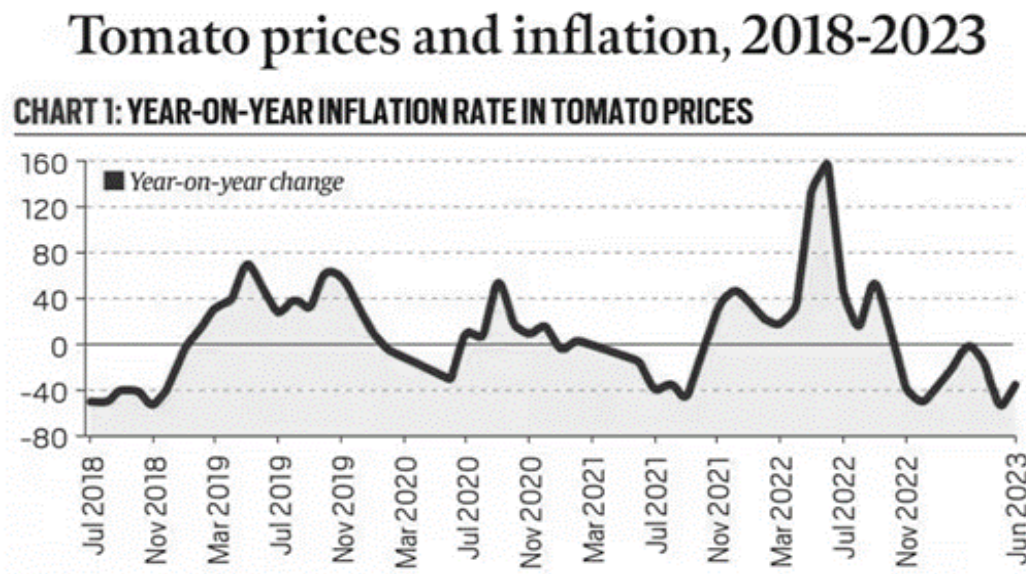


Figure 2. Tomato prices and inflation, 2018-2023 (Drishti IAS, 2023)

Although prior works have examined tomato diseases or the use of technologies such as CNNs, IoT, and UAVs independently, there is no comprehensive study that simultaneously considers the tomato life cycle, stage-specific disease occurrence, and a detailed assessment of emerging technological approaches together. To address the comprehensive study on tomato cultivation, advanced technologies and disease management, the following research questions are considered in this paper:

RQ1: How does the life cycle of tomato plants influence their susceptibility to various diseases?

RQ2: What are the specific varieties of tomatoes cultivated in different regions of India?

RQ3: What environmental factors are essential for optimal tomato crop growth?

RQ4: What are the primary diseases affecting tomato plants and their causes, symptoms, and control methods? How do these vary across the different stages of the plant's life cycle and what are current limitations in disease detection?

RQ5: How effective are various advanced technologies, such as AI, Drones/UAVs, IoT, cloud and edge computing, and blockchain in sustainable tomato cultivation?

The contribution of this paper is to address these research questions through a thorough study of tomato cultivation, disease occurrence, and the role of new technologies in improving sustainable farming.

The paper is organized as shown in Figure 3. Section 2 examines the entire tomato plant life cycle, from seed planting to fruit. Section 3 presents a detailed analysis of different tomato varieties and their characteristics. Section 4 discusses the environmental factors essential for optimal tomato crop growth. Section 5 provides details on various types of diseases affecting tomato plants, their causes, and preventive measures. Section 6 discussed the evaluation of various advanced technologies for identifying diseases in tomato plants, evaluating the accuracy of each technique, and the stages at which diseases manifest, based on various research studies. It also discusses the comparative analysis of different methods for real world deployment along with their advantages and challenges in tomato agriculture. The conclusion and future work are given in Section 7.

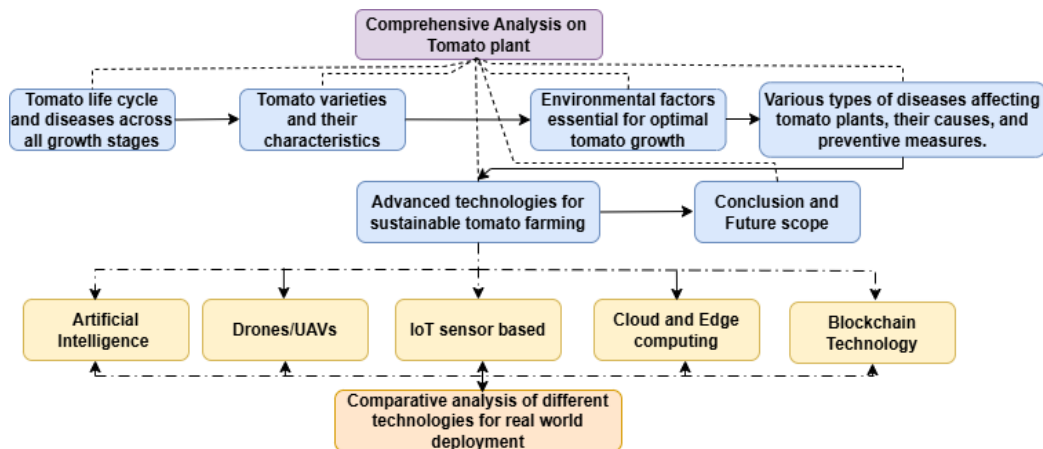


Figure 3. Flowchart detailing the methodology used for the analysis

2. Growth Stages of Tomato Life Cycle

The life cycle of tomatoes is depicted in Figure 4. It starts with small seeds being sown in containers around 6-8 weeks before the expected date of the spring frost. These seeds require moist, well-drained soil and should be sown to a ¼ inch deep. The seeds develop into seedlings first and then develop secondary buds known as cotyledons. These plants bear and thicken their stems and roots; they bear yellow flowers which after pollination yield green tomatoes. These fruits mature to develop a skin and flesh condition that makes them ready for sale. During this cycle, tomatoes require watering, sunlight and protection from pests to maintain ideal growth and production.

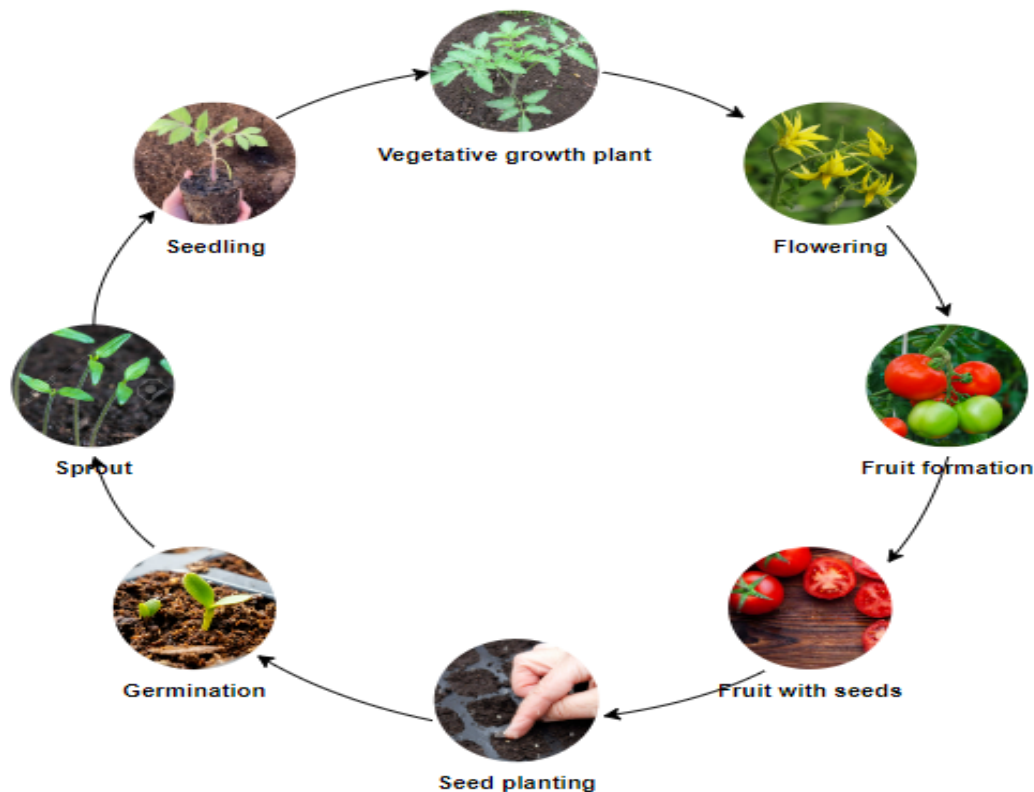


Figure 4. Tomato life cycle

Table 1 shows the entire life cycle of the tomato plant with the duration of each stage of growth in the plant with the diseases observed and their reason for occurrence, indicating a clear idea of the vulnerabilities of stages in tomato plant growth. The data show that tomato plants are prone to diseases at every stage of growth, from seed planting to ripening. Each stage presents unique risks linked to environmental conditions, watering practices, and pathogen presence in the soil or air. While disease management is essential throughout the tomato life cycle, stages 4 to 7 (seedling to fruiting) are most crucial for farmers, as infections during these stages directly affect plant health, fruit development, and final yield.

Table 1. Tomato plant growth stages, duration, common diseases, and causes

Growth Stage	Description	Duration (Approx.)	Common Diseases	Cause	Ref
Seed Planting	Tomato seeds are sown indoors in pots before the last frost. Plant can also be propagated from seedlings, stem cuttings, or grafting. Seeds are sown about ¼ inch deep in moist, well-drained soil.	Week 0 - 1	Seed rot, Damping off	Caused by soil-borne fungi (<i>Pythium</i> , <i>Rhizoctonia</i>) due to overwatering or poor soil drainage.	(Zhou et al., 2021)
Germination	Seeds sprout and form small root-like structures. Keep soil moist; move seedlings to a bright area after germination; water frequently.	Week 1 - 2	Damping off, Seedling blight	High humidity and poor air circulation promote fungal infection and seedling collapse.	(Sultana et al., 2020)
Sprouting	Roots and shoots continue to develop; seedlings are thinned to one per pot. Thin seedlings; maintain moisture and light.	Week 2 - 3	Early seedling wilt, Bacterial blight	Weak root development and excessive moisture create favorable conditions for bacteria and fungi.	(Sultana et al., 2020)
Seedling	Plant has more than two leaves and is ready for transplanting. Transplant outdoors when frost has passed and soil temperature is around 60°F.	Week 3 - 5	Leaf spot, Damping off, Early blight	Overcrowded planting and excessive watering lead to fungal spread and poor aeration.	(Alizadeh-Moghaddam et al., 2024)
Vegetative Growth	The plant focuses on stem and leaf growth. Use water often and balanced fertilizer.	Week 5 - 9	Early blight, Septoria leaf spot, Bacterial speck	Warm, wet weather and overhead irrigation favor spore germination and leaf infection.	(Chakravarthy & Raman, 2020)
Flowering	Flowers appear, indicating plant maturity and readiness to produce fruit. Maintain watering, ensure adequate sunlight and nutrients.	Week 9 - 12	Late blight, Bacterial spot, Tomato leaf curl virus, Powdery Mildew	Prolonged humidity and leaf wetness create ideal conditions for fungal and bacterial proliferation.	(Mazumdar et al., 2021)
Fruiting	Flowers develop into tomatoes. Use water often and fertilizer high in phosphorus to support fruiting.	Week 12 - 16	Buckeye rot, Tomato spotted wilt virus (TSWV), Fusarium wilt	Soil-borne pathogens and insect vectors (thrips) contribute to disease onset and spread.	(Gupta et al., 2022)
Ripening	Fruits change color (green to red/yellow/orange) and become ready for harvest. Maintain watering and harvest ripe fruits carefully from the vine.	Week 16 - 20	Fruit rot, Bacterial canker, Tomato Mosaic Virus (TMV)	Calcium deficiency, irregular irrigation, and fruit surface injury promote infection.	(Cordon et al., 2022)

Tomato cultivation from seed to harvest is normally a process that takes between 70 to 100 days but the time may be adjusted depending on the specific tomato type grown and conditions. Some of the parameters that determine the growth rate of the plant include modes of planting, types of soil, climatic conditions, irrigation and disease control amongst others. Knowledge of these variables is crucial if yield is to have to be maximized, and if tomatoes are to be grown successfully.

3. Tomato Varieties: Cultivation Areas and Specifications

In order to provide a full understanding of the wide range of tomato varieties, detailed data has been compiled from various research papers (Sharma et al., 2019; Blancard, 2012; National Horticulture Board, 2025b). Table 2 includes details regarding the tomato varieties, specific areas of cultivation and the special specifications of each tomato variety. By analyzing these details, researchers and agricultural practitioners can get insights into the optimal growing conditions and characteristics that define each variety (National Horticulture Board, 2025b). This information is essential for selecting appropriate varieties for different climatic regions, improving cultivation practices, and enhancing overall tomato production (Sharma et al., 2019).

Diseases can differ in appearance and frequency due to the connected biologically and physiologically differences of tomato varieties. This fact alone influences the detection performance variability of AI models that are trained on heterogeneous datasets. To ensure that image-based disease detection models detect correctly, tomato differences in leaf morphology, pigment content, canopy density, and disease resistance must be taken into account. Variations in leaf color and texture can change the visual appearance of diseases, thus making model interpretation more difficult, but varieties with dense foliage or overlapping leaves are more likely to have unclear disease symptoms. In addition, hybrid cultivars with partial resistance to some pathogens typically show only a few symptoms, which might lead deep learning systems to misclassify them. These differences raise the importance of using multi-variety datasets and adaptive model training strategies for robustness and generalization in tomato disease detection.

4. Environmental Factors Essential for Optimal Tomato Crop Growth

The environmental factors shown in Figure 5 demonstrate the ideal tomato growing conditions. In general, a temperature range of 20°C to 30°C is perfect for tomato growth. Usually, planting calendars are in line with nature's cycles. For instance, Zaid planting is done in February - March, Kharif planting in June - July and Rabi planting in October - November. Soft, loamy soil with a pH close to 7 that is slightly alkaline is the best soil for tomatoes. The plant must be exposed to 6 to 8 h of direct sunlight per day to remain healthy. As tomatoes need one to one and a half inches of water per week, drip irrigation is mostly chosen to provide frequent but moderate watering. Along with compost or organic manure, NPK fertilizers should be used to enrich the soil. The right and complete crop protection from pests and diseases is also very important. By sustaining these conditions, growers can elevate the tomato crop growth and yield to a great extent (Gleason & Edmunds, 2005; Blancard, 2012; Gangadhar & Sowmya, 2020).

Table 2. Details on tomato varieties

Variety	Area of Cultivation	Specification
	Vaishali	Hot and humid areas
		Determinate hybrid; medium-sized (~100 g) fruits. Resistant to <i>Fusarium</i> and <i>Verticillium</i> wilts; good for juice.
	Rupali	Suitable for various regions
		Determinate, compact, early. Medium-sized (~100 g), round, deep red fruits. Resistant to <i>Fusarium</i> and <i>Verticillium</i> .
	Rashmi	Maharashtra, Karnataka, Andhra Pradesh, and Madhya Pradesh; Kharif and rabi seasons
		Determinate. Round, firm fruits with good color. Resistant to <i>Fusarium</i> and <i>Verticillium</i> ; suitable for processing.
	Rajni	Northern and central India
		Very early, determinate, round red fruits. Performs well in dry and semi-arid regions. Ideal for long-distance transportation.
	Pusa Ruby	Grown across India (all seasons)
		Early, yellow stem end, slightly furrowed, uniform ripening, and processing. Average yield 32.5 t/ha.
	Co 1	Southern India (Tamil Nadu and nearby states)
		Round, yellow stem end, determinate, uniform ripening.

Table 2. Details on tomato varieties (continued)

Variety	Area of Cultivation	Specification
	Sioux	Hilly regions (cool climates)
	Punjab Chuhra	Ludhiana
	ArkaMegha li	Kharif season
	Arka Vishal	Kharif/Rabi season
	Arka Ahuti	All seasons southern and central regions (Karnataka, Tamil Nadu, Andhra Pradesh, Maharashtra)

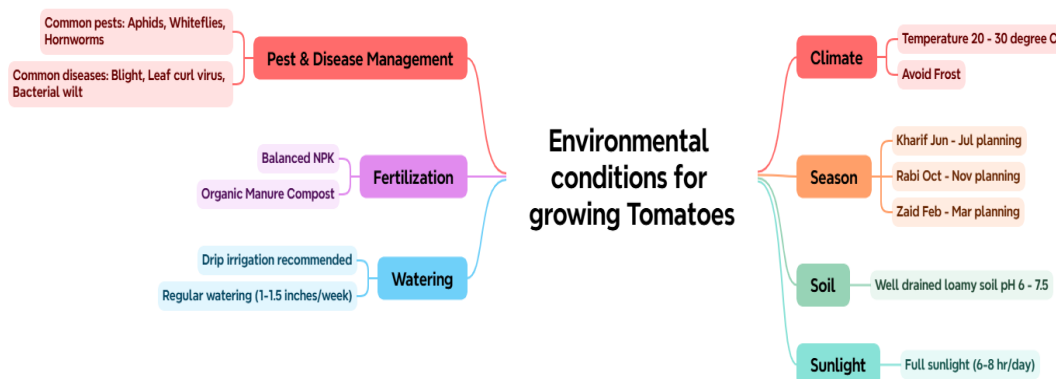


Figure 5. Environmental factors for growing tomatoes

Various environmental factors influence the severity of different diseases as well as the precision of their diagnosis. Variations in temperature and humidity have a significant impact on the growth of pathogens such as *Alternaria solani*, which causes early blight, and *Phytophthora infestans*, which causes late blight. Both of these pathogens multiply more in environments with high humidity and moderate temperatures (20-25°C). Excess moisture due to over-irrigation or poor drainage provides excellent conditions for bacterial wilt and damping-off especially during the seedling stage. Conversely, extreme heat stress and low humidity can induce physiological disorders that mimic disease symptoms, thereby confusing vision-based detection systems. Light intensity and duration also impact the spectral reflectance of leaves, influencing the performance of image-based classification models under natural versus controlled illumination. Similarly, soil pH and nutrient imbalances alter plant physiology and leaf coloration, which can lead to false positives or misclassifications in automated disease detection frameworks (Sun et al., 2025).

5. Tomato Plant Diseases and Its Control Methods

To provide a comprehensive understanding of how diseases impact tomato plants, a Table was developed using information compiled from various research articles. Table 3 contains information on diseases by highlighting the characteristic symptoms of each disease and provides general control measures (Vasudeva & Sam Raj, 1948; National Horticulture Board, 2025a).

Table 3. Disease details along with symptoms and control methods

Ref	Disease	Symptom	Control Method
(Sultana et al., 2020; Zhou et al., 2021)	Damping Off (<i>Pythium aphanidermatum</i>) 	Seedling death before or after emergence, water-soaked stem at soil line	Seed treatment with beneficial fungi (<i>Trichoderma viride</i>) or fungicide (Thiram), drenching seedlings with fungicide (Dithane M45)
(Chakravarthy & Raman, 2020; Rahman et al., 2023; Alizadeh-Moghaddam et al., 2024)	Early Blight (<i>Alternaria solani</i>) 	Seedling: Small, black lesions on leaves * Plant with fruit: Larger lesions with concentric rings on leaves and fruits, can affect ripening	Remove infected plant parts, crop rotation, fungicides (Difolatan, Dithane M-45, Bavistin)
(Gupta et al., 2022)	Buckeye Rot (<i>Phytophthora parasitica</i>) 	Grayish-green or brown water-soaked spots on fruits near the ground, with concentric rings	Good drainage, staking plants, removing lower foliage and fruits, fungicide (Difolatan)
(Zhou et al., 2021; Mazumdar et al., 2021; Singh et al., 2022b; Rahman et al., 2023)	Late Blight (<i>Phytophthora infestans</i>) 	Plant with fruit: Large, greenish-black, water-soaked lesions on leaves and fruits, white mold on undersides of leaves in humid conditions	Crop rotation, disease-free seeds and transplants, fungicides (Captafol, Dithane M-45)
(Zhang et al., 2021; Devi et al., 2022; Ramirez et al., 2022).	(<i>Fusarium oxysporum</i> f. sp. <i>lycopersici</i>) 	Plant with fruit: Yellowing of veins, chlorosis of leaves, wilting and death of leaves, browning of vascular system, stunted plant	Remove infected plants, drench nursery beds with fungicide (Carbendazim), seed treatment with fungicide (Thiram), and crop rotation

Table 3. Disease details along with symptoms and control methods (continued)

Ref	Disease	Symptom	Control Method
(Mani et al., 2021; Rahman et al., 2023)	Septoria Leaf Spot (<i>Septoria lycopersici</i>) 	Plant with fruit: Numerous, small, gray, circular leaf spots with dark borders	Remove infected plant parts, seed treatment with fungicides (Thiram, Dithane M-45), fungicide (Dithane Z-78)
(Pramanik et al., 2021; Bhatia et al., 2022)	Powdery Mildew (<i>Leveillula taurica</i>) 	Plant with Fruit: White powdery coating on leaf surfaces, stunted and distorted leaves	Fungicides (Karathane, Wettable Sulfur)
(Chou et al., 2022; Singh et al., 2022a)	Bacterial Wilt (<i>Pseudomonas solanacearum</i>) 	Rapid and complete wilting of plants, white bacterial ooze from cut stems	Crop rotation, seed treatment with antibiotics (Streptomycin)
(Giri et al., 2023; Qiao et al., 2020)	Bacterial Leaf Spot (<i>Xanthomonas campestris</i> pv. <i>vesicatoria</i>) 	Plant with Fruit: Small, brown, water-soaked spots on leaves and fruits	Disease-free seeds and seedlings, crop rotation, copper and organic fungicide sprays, antibiotic spray (Agrimycin-100)
(Cordon et al., 2022; Tripathi et al., 2022b)	Bacterial Canker (<i>Clavibacter michiganensis</i> pv. <i>michiganensis</i>) 	Plant with Fruit: Wilting of leaves, light streaks on stems and petioles, cankers on stems, water-soaked spots with halos on fruits	Hot water seed treatment, seed treatment with fungicide (mercuric chloride), crop rotation, seed soaking in antibiotic solution (Streptomycin)

Table 3. Disease details along with symptoms and control methods (continued)

Ref	Disease	Symptom	Control Method
(Elsharkawy et al., 2022)	Tomato Mosaic Virus (TMV) 	Plant with Fruit: Light and dark green mottling on leaves, wilting, distorted and stunted leaves, reduced fruit quality	Disease-free seeds, seed treatment with trisodium phosphate, removing infected plants, crop rotation
(Dhaliwal et al., 2020)	Tomato Leaf Curl Virus (TLCV) 	Plant with Fruit: Severe stunting, downward curling and rolling of leaves, leathery and brittle older leaves, bushy appearance	Remove infected plants, remove alternate hosts, control whitefly population with insecticides (Furadan, Dimethoate, Monocrotophos), use of border crops or polythene mulch
(Qi et al., 2022)	Tomato Spotted Wilt Virus (TSWV) 	Plant with Fruit: Numerous small, dark spots on leaves, bronzed leaves, spotted fruits	Remove infected plants, manage thrips populations, use resistant plant varieties, and practice crop rotation.

6. Evaluation of Various Advanced Technologies for Sustainable Tomato Farming

As the demand for agricultural products keeps growing, new technology can help to guarantee better yield and remain competitive in the agricultural sector. Hence, relying on advanced technologies such as Artificial Intelligence (AI), Internet of Things (IoT), Drones/UAVs, Cloud computing and Blockchain are necessary. Beyond enabling early and accurate disease detection, these technologies facilitate resource management, leading to more efficient and sustainable practices in tomato farming. In this section we bring the spotlight to this application of these advanced methods and the results they have on tomato crop cultivation.

6.1 Artificial intelligence

AI including ML & DL are revolutionizing the farming of tomatoes by assisting in early detection and control of diseases, which helps in improving the yield and quality. Convolutional Neural Networks (CNNs) (Yoren & Suyanto, 2021) can be used to analyze images of tomato plants to detect diseases such as early blight, late blight, leaf mold, and bacterial spot at various stages of growth. Transfer learning using pre-trained models like VGG16, ResNet and Inception allows for improved accuracy, especially when there is limited data. Recent methods integrate several models or types of data to increase performance. These include hybrid models (CNN + Vision Transformer), ensemble models (ResNet50 + MobileNetV2) (Tarek et al., 2022; Tripathi et al., 2022a), and multimodal approaches that take into account images and information about the environment. Such DL techniques assist farmers to keep a check on the health of the plants, resources should be managed efficiently, crop losses should be prevented, thus making tomato cultivation more productive and sustainable. Table 4 gives details on multiple model types present in AI technique (Luna-Benoso et al., 2021; Rani et al., 2024).

6.1.1 Convolutional neural networks (CNNs)

Convolutional Neural Networks offer an efficient and accurate solution for detecting and managing diseases in tomato plants, addressing a critical issue for this crop. CNNs are particularly well-suited for processing image-based inputs, making them ideal for agricultural applications where visual symptoms are a primary indicator of disease (Tripathi et al., 2022a). By utilizing high-resolution images captured by drones or satellites, CNNs can accurately identify various tomato plant diseases such as late blight, early blight, and leaf mold, at early stages. This early detection capability is crucial for timely intervention and disease management, which can significantly reduce crop losses and improve yield.

Figure 6 shows the data flow and architecture of convolutional neural network (CNN) (Tripathi et al., 2022a) that was used to detect tomato diseases using images. The CNN takes RGB- tomato leaf images, which are scaled to 224×224 and augmented with additional 3 dimensions and then normalized with ImageNet mean variance statistics so that they can converge consistently during training. The model involves the use of several convolutional layers using 3×3 kernels with a stride of 1, and, finally, the use of Rectified Linear Unit (ReLU) activation functions to provide non-linearity. Feature maps generated by convolution operations are progressively downsampled using 2×2 max-pooling layers, which reduce spatial dimensionality while preserving discriminative visual features such as lesion boundaries, chlorosis patterns, and necrotic spots. Mathematically, the extraction of the feature in the l th layer can be written as follows:

$$F_l = \sigma(W_l * F_{l-1} + b_l) \quad (1)$$

where $*$ denotes convolution and $\sigma(\cdot)$ represents the ReLU activation in equation 1. The features extracted are flattened and fed to fully connected layers, after which they are fed to a Softmax classifier that gives the probability of diseases in a given set of preset classes (e.g., Early Blight, Late Blight, Leaf Mold, Healthy). Training of the model is done offline with categorical cross-entropy loss on labeled datasets and Adam optimizer ($\eta=1 \times 10^{-4}$), and inference is done in real-time on either a cloud or edge platform based on the availability of connectivity (Hassan & Maji, 2022; Verma et al., 2020).

Table 4. Details on different model types

Category	Technique / Model Type	Description
Machine Learning (ML) Models	Support Vector Machine (SVM)	Classifies tomato leaf diseases using handcrafted features (e.g., color, texture, and shape) extracted from images; effective for small datasets.
	Random Forest (RF)	Ensemble-based classifier using multiple decision trees to enhance prediction accuracy and robustness for disease detection.
	K-Nearest Neighbor (KNN)	Classifies diseases based on similarity between new input features and labeled training samples; simple but less scalable.
	Naïve Bayes	Probabilistic model for classifying tomato diseases using statistical relationships among features.
	Decision Tree (DT)	Rule-based model used to classify disease categories based on segmented and extracted image features.
Deep Learning (DL) Models	Convolutional Neural Networks (CNNs)	Used for image-based disease detection by analyzing leaf patterns and spots. Examples: Detecting early blight, late blight, leaf mold, and bacterial spot.
	Transfer Learning (e.g., VGG16, ResNet50, InceptionV3, MobileNetV2))	Pre-trained on large datasets (like ImageNet) and fine-tuned for tomato disease images. Useful when training data is limited.
	Hybrid Models (CNN + Vision Transformer)	Combine CNN's local feature extraction with transformer's global attention to improve classification accuracy.
	Ensemble models (Combination of multiple CNNs e.g., ResNet50 + MobileNetV2)	Aggregate predictions from multiple models to enhance robustness and reduce individual model bias.
	Multimodal / Data-Fusion models	Integrate visual and environmental features to improve disease prediction accuracy and contextual understanding.

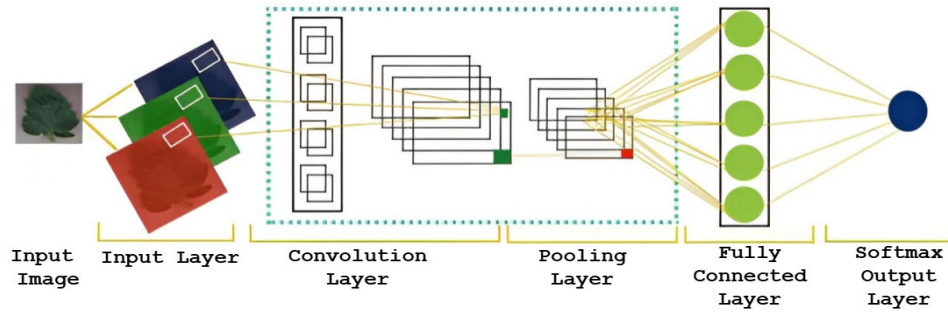


Figure 6. CNN architecture (Kibriya et al., 2021)

A thorough analysis of numerous research studies that have used a variety of AI techniques to detect tomato plant diseases can be found in detail shown in Table 5. The Table compiles data from several publications and groups them according to the diseases under consideration, the technology employed, the accuracy rates, and the severity of the plant disease. By providing farmers with a potent tool to improve crop health and productivity, CNN integration into agricultural practices marks a substantial advancement in the management of tomato plant diseases.

6.1.2 Transfer learning

The VGG16-based transfer learning model used to enhance tomato disease classification performance in the case of scarce labelling information, is shown in Figure 7. VGG16 comprises a deep hierarchy of convolutional layers organized into five convolutional blocks, each using 3×3 convolutional kernels, ReLU activation $f(x)=\max(0,x)$ and max-pooling operations, followed by fully connected and Softmax layers. The convolutional feature extraction algorithm can be stated as:

$$F_k = W_k * X + b_k, \quad (2)$$

where X denotes the input feature map and W_k represents the learned filter weights in equation 2.

In this work, VGG16 is initialized with weights pre-trained on the ImageNet dataset, enabling the reuse of generic visual representations such as edges, textures, and shapes. In the fine-tuning stage, previous convolutional networks are frozen such that they do not lose any general feature extraction ability, but instead higher level networks are retrained with tomato disease images to acquire crop-learned visual features. The last classification is done using the Softmax, which gives the probability of the classes that are associated with tomato disease classes. This transfer learning approach reduces overfitting and solves the problem of dataset scarcity that is usually faced in agricultural imaging. Data augmentation in terms of rotation, flipping, illumination adjustment is used to conduct training to enhance generalization. In related studies, The inference stage achieved accuracy ranging from 92% and 99%, demonstrating a higher level of robustness than shallow CNNs when subjected to variation in lighting conditions, background noise and leaf orientation (Kibriya et al., 2021). The developments of recently developed tomato disease

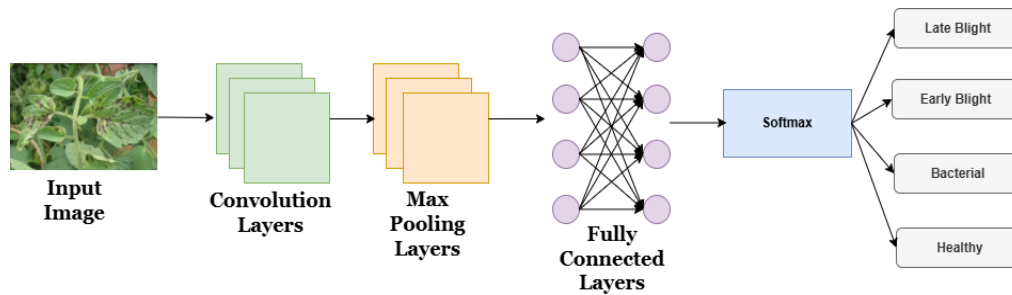


Figure 7. VGG architecture

detection technologies are not solely based on the size of the dataset, but also on the decisions made by the algorithm in the machine learning and deep learning models. In convolutional neural networks (CNNs), commonly adopted architectures such as VGG, ResNet, MobileNet, and Inception differ in depth, kernel configurations, and parameter efficiency. Shallow CNNs with small kernel sizes (e.g., 3×3) are effective for capturing local lesion patterns, while deeper architectures leverage hierarchical feature learning to model complex disease textures. Residual connections in ResNet architectures mitigate vanishing gradient issues and enable stable training of deeper networks, resulting in improved generalization. Lightweight models, including MobileNet and SqueezeNet, employ depthwise separable convolutions and parameter reduction strategies to minimize computational overhead, making them suitable for edge and mobile deployment without substantial loss of accuracy. Transfer learning is also very popular, where models already trained on large-size datasets like ImageNet are further fine-tuned to tomato disease datasets which saves training time and enhances convergence to limited conditions of the labeled data. Some of the typical optimization techniques are data augmentation, to deal with class imbalance, dropout and batch normalization, to avoid overfitting, and adaptive optimizers including Adam. Beyond single-model architecture, recent studies explore hybrid and ensemble approaches, combining CNNs with transformers, recurrent models, or classical classifiers to capture both local visual features and global contextual information. Multimodal fusion methods integrate image data with environmental parameters (e.g., temperature, humidity, soil moisture), either through feature-level concatenation or decision-level fusion, leading to more robust disease prediction under real-field variability. All these algorithmic decisions are made to achieve a better representation of accuracy, scale and deployment in a practical tomato farming context. Table 5 gives the summary of the related work with respect to transfer learning with the parameters like diseases considered, technology used, accuracy, and levels of infection, TBS (Tomato Bacterial Spot), TEB (Tomato Early Blight), TLB (Tomato Late Blight), TLM (Tomato Leaf Mold), TSL (Tomato Septoria Leaf Spot), TSM (Tomato Spider Mites), TTS (Tomato Target Spot), TYLCV (Tomato Yellow Leaf Curl Virus), TMV (Tomato Mosaic Virus), and TH (Tomato Healthy).

Table 5. Applications of AI methods for tomato farming

Ref	Disease Considered	Technology Used	Accuracy	Levels (Tomato Life Cycle)	Outcome
(Bhatia et al., 2022)	Spider mite, Powdery mildew	Multi-Layer Perceptron (MLP), Radial Basis Function (RBF) based Artificial Neural Network (ANNs), Learning Vector Quantization (LVQ)	98.65%	Seedling, Vegetative, Flowering, Fruiting	ANN-based models outperform handcrafted feature classifiers
(Luna-Benoso et al., 2021)	Tomato mosaic virus, Septoria leaf spot, Late blight,	Support Vector Machines (SVM) classifiers, Multilayer Perceptron (MLP), K Nearest Neighbors (K-NN)	93.50%	Fruiting, Vegetative, Flowering	Classical ML effective but sensitive to feature quality
(Islam, 2020)	Early blight, late blight, bacterial spot, leaf mold	VGG-19 and ResNet (CNN)	96.30%	Seedling, Vegetative, Flowering, Fruiting	Deep CNNs improve robustness over shallow models
(Bala et al., 2025)	TBS, TEB, TLB, TLM, TSL, TSM, TTS, TYLCV, TMV, TH	MobileNetV2	99.46%	Seedling, Vegetative, Flowering, Fruiting	Lightweight CNN suitable for mobile/edge deployment
(Nyasulu et al., 2023)	Tomato fungal leaf diseases	Support Vector Machines (SVM)	94%	Seedling, Vegetative, Flowering, Fruiting	Optimized CNN balances accuracy and computational cost
(Tarek et al., 2022)	TH, Bacterial Spot, TEB, TLB, Septoria, Target Spot, TSM, TYLCV, TMV, TLM	AlexNet	Accuracy: 96.68% Precision: 96.5% Recall: 96.5%	Seedling, Vegetative, Flowering, Fruiting	Early CNN architecture improves accuracy compared to traditional ML
(Zhou et al., 2022)	TH, Bacterial Spot, TEB, Late Blight, TSL, Target Spot, Two-spotted TSM, TYLCV, TMV, TLM	MobileNetV3	Accuracy: 98.25% Precision: 98.2% Recall: 98.2%	Seedling, Vegetative, Flowering, Fruiting	Optimized MobileNet balances accuracy and efficiency for edge deployment
(Vallejo-Pérez et al., 2021)	Bacterial canker of tomato	PCA + MLP (Principal Component Analysis combined with Multilayer Perceptron)	99%	Seedling, Vegetative, Fruiting	Dimensionality reduction enhances ANN performance and convergence speed

Table 5. Applications of AI methods for tomato farming (continued)

Ref	Disease Considered	Technology Used	Accuracy	Levels (Tomato Life Cycle)	Outcome
(Ouhami et al., 2020)	TBS, TEB, TLB, TLM, TSL, TSM, TTS, TYLCV, TMV, TH	Improved MobileNetV3	98.25%	Seedling, Vegetative, Flowering, Fruiting	Increase robustness for multi-disease classification
(Medici et al., 2020)	TH, Bacterial Spot, TEB, TLB, Septoria, Target Spot, TSM, TYLCV, TMV, TLM	ResNet50 CNN Model	99.74%	Seedling, Vegetative, Flowering, Fruiting	Deep residual learning achieves very high accuracy for multi-class detection
(Vandana & Goel, 2021)	Tomato Bacterial Spot, Tomato Early Blight, Tomato Late Blight, Tomato Target Spot, Tomato Yellow Leaf Curl, Tomato Mosaic Virus, Tomato Septoria Leaf Spot	Convolutional Neural Network (CNN), Decision Trees (DT), K-Nearest Neighbor (KNN), Random Forest Classifier (RFC), Naïve Bayes (NB), Support Vector Machines (SVM)	94.83%, 15.97%, 96.54%, 99.93%, 94.8%, 96.87%	Seedling, Vegetative, Flowering, Fruiting	Ensemble and CNN models outperform traditional classifiers
(Yoren & Suyanto, 2021)	Alternaria Solani, Septoria Leaf Spot, Yellow Virus	CNN	97.80%	Vegetative, Flowering	CNN achieves strong performance for limited disease classes
(Aversano et al., 2020)	Bacterial Spot, Early Blight, Healthy, Late Blight, Leaf Mold, Septoria Leaf spot, Target Spot, Mosaic Virus, Yellow Leaf Curl Virus, Two spotted spider mite	VGG-19, Xception and ResNet-50	97%, 95%, 60%	Seedling, Vegetative, Flowering, Fruiting	Performance varies across architectures, highlighting model sensitivity
(Hasan et al., 2019)	Bacterial Spot, Early Blight, Healthy, Late Blight, Leaf Mold, Two spotted spider mite, Target Spot, Mosaic Virus, Yellow Leaf Curl Virus, Septoria Leaf spot	Inception v3	99%	Seedling, Vegetative, Flowering, Fruiting	Inception architecture effectively captures multi-scale disease features

Table 5. Applications of AI methods for tomato farming (continued)

Ref	Disease Considered	Technology Used	Accuracy	Levels (Tomato Life Cycle)	Outcome
(Alzahrani, 2025)	Buckeye rot, Tomato Mosaic Virus	Fusion of VGG16 + ResNet-50 features; hybrid classifiers	Accuracy: 97% Precision: 90% Recall: 80%	Fruiting	Feature fusion improves discrimination between visually similar diseases
(Mezenner et al., 2025)	Early blight, Late Blight, Septoria Leaf spot	Vision Transformer + CNN hybrid (compact convolutional ViT)	98.90%	Leaf stage	Transformer-based models capture global contextual disease patterns
(Zhang et al., 2022)	Blight, Leaf mold, Bacterial spot, Mosaic virus	Multimodal fusion (images + environmental parameters), transfer learning + ResNet34 backbone	99.30%	Leaf stage	Fusion of image and environmental data improves accuracy
(Sun et al., 2025)	Early blight, Late blight, Bacterial spot, Mosaic virus	Hybrid CNN-Transformer (EfficientNetV2 + Swin Transformer) combining local & global features	Accuracy: 99.7% Precision: 93.1% Recall: 95.2%	Vegetative, Flowering, Fruiting	Hybrid architecture captures local and global features
(Sharma et al., 2025)	Early blight, Late blight, Bacterial spot, Mosaic virus	Ensemble model (ResNet50 and MobileNetV2)	Accuracy: 99.91% Precision: 99.92% Recall: 99.9%	Leaf stage	Ensemble learning reduces misclassification and improves stability
(Islam et al., 2022)	Early blight, late blight, leaf mold, septoria leaf spot, bacterial spot, target spot, mosaic virus, spider mites	Multimodal hybrid deep learning with attention-based dilated CNN feature extractor + logistic regression classifier	99.72%	Leaf stage	Attention improves discriminative feature learning

Table 5 shows that there is a definite shift in the usage of the traditional machine learning models to deep and hybrid learning models in detecting tomato disease. Although classifiers like SVM, KNN, and MLP delivered higher accuracy in the initial stages of classifier development, deep CNNs (e.g., VGG or ResNet or MobileNet or Inception) are able to provide better accuracy and generalization through various growth phases. The recent hybrid, ensemble and multimodal models go a step further to make it robust under real field conditions and are more scalable and deployable. Recent research indicates that there has been a clear movement towards standalone CNN-based models to hybrid, ensemble and multimodal architecture, which are proved to be more accurate (usually over 99%) at detecting tomato disease. The hybrid models are superior to the traditional CNNs in that they combine local feature representation (CNNs) with global contextual knowledge (transformer attention), and thus they are more effective in distinguishing visually similar diseases as experiences in complex fields. Ensemble methods also enhance performance through combining the predictions of several different models. This helps in minimizing bias and improving resistance to noise and change in illumination, and background clutter, which are frequent features of agricultural images. Multiform techniques combining leaf image with environmental variables like temperature, humidity and soil moisture also demonstrated extra accuracy, since there is a close relationship between disease expression and growing conditions. Transfer learning is also an important aspect, as it facilitates the acquisition of visual representations and enables high accuracy by using low-volume agricultural data. Such tendencies imply that such systems will evolve next to be less reliant on reactive image-based classification but context-aware, multimodal, and hybrid AI. Edge-compatible design lightweight models and model compression technologies are required to trade off accuracy with latency, scalability, and rural connectivity limits. Despite these many approaches, current tomato disease detection systems still face several limitations. Many models struggle with generalization across different tomato varieties, environmental conditions, and imaging devices. There is also a lack of models designed to address all stages of the tomato growth cycle, leading to gaps in early-stage disease detection and lifecycle-based monitoring.

6.2 Drones and UAVs

Tomato diseases are well addressed by the use of drones or Unmanned Aerial Vehicles (UAVs) in recent studies. UAVs present a fast method of observing vast areas of the field at once and giving the best aerial perspective of the crop. These are equipped with high resolution cameras. They are able to rapidly take photos of tomato fields, which help in identifying plants that are affected by diseases at early stages (Shahi et al., 2023; Zhong et al., 2025). Also, UAVs can be equipped with various sensors and imaging systems including multispectral cameras, thermal sensors, LiDAR to gather facts for disease mapping and spatial analysis so that the correct control of affected zones is possible (Yallappa et al., 2024). Precision spraying systems when fitted in drones help to optimize fertilizer utilization in tomato farms by spreading fertilizer evenly and efficiently in the field and hence minimizing wastage of both money and fertilizer. Inspection through high-resolution cameras and multispectral sensors enables the farmer to determine nutrient deficiencies and disease-infected flowers from a distance. These instant remarks enable more rational decision making in crop management. Drone-produced aerial maps that capture crop conditions and development bring critical differences into focus and can be added to soil sensors and weather reports for crop management strategies (Shahi et al., 2023). Drone types and their applications in tomato cultivation are tabulated in Table 6 (Egi et al., 2022; Chin et al., 2023; Chavez-Martinez et al., 2025; Zhang et al., 2025).

Table 6. Drone types and their applications in tomato cultivation

Drone Type	Example Model	Applications in Tomato Cultivation
Multispectral Drones (Zhang et al., 2025)	DJI Phantom 4 Multispectral, Parrot Sequoia+	Monitoring plant health, detecting stress, and analyzing NDVI (Normalized Difference Vegetation Index) for early disease detection.
RGB Camera Drones (Chavez-Martinez et al., 2025)	DJI Mavic 3, DJI Mini 4 Pro, Autel EVO II	Capturing high-resolution field images for growth monitoring, canopy coverage, and general inspection.
Thermal Imaging Drones (Chin et al., 2023)	DJI Matrice 300 RTK + Zenmuse H20T, FLIR-equipped drones	Assessing soil moisture, irrigation efficiency, and identifying water stress in tomato plants.
Spraying Drones (Chavez-Martinez et al., 2025)	DJI Agras T30, XAG P30, Hylío AG-272	Precision pesticide, fungicide, and nutrient spraying to control diseases and improve yield.
Mapping / Survey Drones (Egi et al., 2022)	SenseFly eBee X, WingtraOne	Creating 2D/3D field maps for planning, crop area estimation, and precision farming.
Fixed-Wing Drones (Egi et al., 2022)	SenseFly eBee SQ, Delair UX11	Covering large tomato farms efficiently for aerial surveys and long-range monitoring.

Moreover, drones also reduce the effects on the environment as they can be used to apply fertilizers and pesticides to specific areas and in amounts that would not pollute water and harm communities of living beings. This sustainable exercise leads to having healthier soils and salubrious wild life. Furthermore, the use of drone can reduce the burden of large areas rapidly and accurately without relying on human labor and resources (Medici et al., 2020). In general, drones improve productivity, sustainability and performance in tomato farming. Table 7 contains information on diseases considered in tomato farming, and the application of technology.

Table 7 highlights the effectiveness of UAV-based systems in enabling large-scale and non-invasive tomato disease monitoring. Early UAV implementations relied on RGB imagery combined with classical machine learning, while recent studies increasingly adopt multispectral and hyperspectral sensing integrated with deep and temporal learning models. These advances significantly improve early disease detection, spatial coverage, and robustness under real-field conditions, making UAV-based platforms particularly suitable for mid- and large-scale tomato farming.

Table 7. Applications of drone/UAV based systems for tomato farming

Ref	Disease considered	Technology used	Outcome
(Thalluri et al., 2021)	Leaf Mold, Bacterial spot, Late Blight	UAV with Raspberry Pi Camera + ML (Feature Extraction & Edge Detection)	Edge-based image processing enables on-board disease detection with reduced latency
(Kouadio et al., 2023)	Bacterial spot, Late blight, Early blight	UAV Remote Sensing (RGB, Multispectral, Hyperspectral) + ML/DL	Multispectral and hyperspectral sensing improves early disease discrimination
(Narimani et al., 2025)	Bacterial spot, Leaf Mold, Early blight, Branched broomrape (Phelipanche ramosa)	UAV Multispectral Imaging + LSTM + SMOTE. Achieved 88.37 % overall accuracy and 95.37 % recall	Temporal modeling with LSTM enhances recall for imbalanced field datasets
(Abdulridha et al., 2020)	Bacterial Spot (BS), Target Spot (TS), and Tomato Yellow Leaf Curl Virus (TYLCV)	UAV Hyperspectral Sensing + STDA + RBF. Achieved up to 100 % classification accuracy for TYLCV, 99 % for TS, and 98 % for BS in field conditions	Hyperspectral features achieve very high classification accuracy under field conditions

6.3 Sensor and IoT systems

Technology and the smart use of instruments such as sensors and the Internet of Things (IoT) is changing tomato farming by offering accurate and timely information for decision making. Temperature, humidity and soil moisture are controlled through environmental sensing technologies, while via smart irrigation systems this data is harnessed to enhance the watering systems. Soil nutrient sensors and automated fertigation systems enhance orderly nutrient management (Choi et al., 2013). IoT systems are pest detection devices and disease prediction models that help farmers prevent crop losses. Drone and other remote sensing devices give an aerial view of the crops that can be used to monitor crops effectively (Hemming et al., 2020). IoT platforms gather and transmit data where insights and possibly predictive analytics become meaningful (Rezvani et al., 2020). Their use leads to higher production, optimal utilization of inputs and resources, prevention of waste, better quality tomatoes and enhanced food production practices (Goap et al., 2018).

Remote IoT sensors can be useful in present-day tomato farming by affording accurate and real-time information on diverse environmental parameters. Soil pH is important for the proper functioning of useful nutrients as well as disease control and is these days often measured using IoT sensors. These sensors also measure rain to ensure the farmers manage their irrigation timetables to avoid wasting water. Further, improved capacity in weather forecasting helps farmers to prepare for the negative conditions in order to guard their plants from vices of weather extremes.

In addition to these, IoT sensors enhance the general operations of a farm through incorporation with smart irrigation techniques for the conservation of water, hence cutting on expenses and environmental effects. Moreover, the sensors allow for easy auto fertigation which is essential for delivering nutrient inputs to the plant at the optimum periods and at the optimum quantities needed for improving plant health and increasing yields. Pest detecting systems and disease prognosis systems allow identifying a problem at an early stage. This can prevent massive crop losses and the excess use of chemicals. IoT technology for farming makes it possible to offer specific suggestions and information that ensures sustainability within farming, increased quality of tomatoes, and increased agricultural yields. Table 8 shows information on references, diseases considered, and technology used.

Table 8 demonstrates that IoT-based systems primarily enhance tomato disease management by enabling continuous environmental monitoring and real-time decision support. Early studies focused on standalone sensor networks, while recent approaches integrate deep learning, edge computing, UAVs, and robotic platforms to improve detection accuracy and responsiveness. These systems significantly contribute to early warning, precision resource management, and scalable deployment across diverse farming conditions. Using the suggested framework, IoT-edge-cloud pipeline can be utilized to monitor intelligent tomato crops and manage diseases. IoT sensors are installed to monitor the important environmental and soil parameters such as air temperature, relative humidity, intensity of light, moisture content of the soil, and soil pH, all of which are essential factors affecting the progression of diseases. The acquisition of tomato leaf images to produce visual disease analysis is done using RGB imaging devices to capture tomato leaf images at specific growth stages or in the case of detecting abnormal conditions. Sensing data is gathered at periodic short intervals to aid near-real time monitoring, whereas image data is gathered periodically or on demand. Low-power wireless communication technologies (LoRa/LoRaWAN to transmit scalar sensor data, while Wi-Fi or cellular networks are used to transmit image and gateway-to-cloud data) are used to transmit data of field devices. To lower the latency and bandwidth usage, a mixed edge-cloud strategy is embraced. Edge devices are used to do initial data filtering, threshold-based alerts and lightweight inferences with optimized deep learning models, and cloud platforms to store large-scale data, performing more sophisticated analytics, model training and multi-sensor data fusion. This task separation allows scalable low-latency and deployment-ready smart tomato farms.

Table 8. Applications of IoT based systems for tomato farming

Ref	Disease Considered	Technology Used	Outcome
(Choi et al., 2013)	Early blight, Bacterial spot, Late blight	Frequency Domain Reflectometry (FDR) sensors for soil moisture and disease monitoring	Soil moisture sensing enables early disease risk assessment and irrigation control
(Rezvani et al., 2020)	TH, Bacterial Spot, TEB, TLB, Septoria, Target Spot, TSM, TYLCV, TMV, TLM	Wireless sensor nodes with LoRaWAN, multi-channel gateway, MATLAB Simulink-based offline data analysis, and web dashboard for monitoring	LoRaWAN-based IoT enables scalable, real-time field monitoring and alerts
(Goap et al., 2018)	Late Blight, Tomato Yellow Leaf Curl Virus (TYLCV), Powdery Mildew	IoT-based wireless sensor networks	Early identification of disease-conducive conditions
(Darmawan et al., 2021)	Early/Late Blight	Transfer learning using SSD MobileNet V2 integrated with IoT-based data acquisition	Edge-integrated deep learning enables near real-time disease detection
(Chung et al., 2022)	Spider Mites, Early Blight, Septoria Leaf Spot, Leaf Mold, Late Blight, Bacterial Spot, Mosaic Virus, Yellow Leaf Curl Virus, Target Spot	Transfer learning on AlexNet integrated with IoT platform for real-time detection	IoT–CNN integration improves continuous monitoring and detection accuracy
(Verma et al., 2018)	Early Blight, Late Blight, Tomato Yellow Leaf Curl Virus, Tomato Mosaic Virus, Septoria Spot, Target Spot, Bacterial Spot, Leaf Mold, Spider Mites, Powdery Mildew	EN Bloodhound ST214 sensors, temperature & humidity sensors, wireless communication modules, microcontroller-based IoT platform	Environmental sensing enhances disease forecasting and preventive action

6.4 Cloud and edge computing

Cloud computing allows streamlined data gathering, storage, and analysis of the huge volume of data generated by different sensors installed in tomato farms. The data covers aspects like temperature, humidity, nutrient levels, soil moisture, and pest activity. The powerful analytics tools provided by cloud platforms take this data to process it and produce insightful findings, forecast trends, and improve farming methods. By localizing the data processing at the source, for instance, on sensors and Internet of Things devices in the field, edge computing is still supportive of cloud computing (Rudramurthy et al., 2018). For operations that are extremely time-sensitive such as pest control and irrigation, this reduction in latency makes real-time decision-making possible. Edge devices can, therefore, ensure that there are always timely responses to changing conditions, for instance, by being able to instantly trigger pest control if a certain level of pest is exceeded or starting the irrigation system when soil moisture is low (Xia et al., 2021).

To assess various approaches for detecting tomato plant diseases with cloud and edge computing, a comprehensive Table 9 was constructed. The Table offers important insights into the efficacy of various ML and DL techniques in disease detection by methodically arranging the data. In order to improve the general health and productivity of tomato crops and advance the field of plant disease management, it is imperative that the most accurate and dependable techniques be found through this comparative analysis (Sagheer et al., 2020; Devaraj et al., 2024).

Figure 8 presents a cloud and IoT-edge integrated system to monitor tomato diseases in real time and provide advice (Islam et al., 2022). IoT sensors used in the field to gather environmental data include soil moisture, pH, temperature, humidity, rainfall and wind. These sensors are configured to work at a set sampling rate (usually 5-15 min) and send data via lightweight communication protocols like MQTT or HTTP over cellular networks or LoRaWAN. At the same time, leaf images are taken with the help of mobile devices or camera modules and delivered to the cloud where they are preprocessed, that is, resized, normalized, and denoised. Cloud hosted deep learning models can be used to extract features and classify diseases, whilst lightweight rule based checks and threshold alerts can be run at the edge to reduce latency. As a way of maintaining reliability and trust, the derived diagnosis of the disease and treatment recommendations are optionally checked with agricultural specialists before being disseminated. The final advisory including disease type, pesticide recommendations, and preventive measures is delivered to farmers through mobile applications or voice interfaces. For real-world feasibility, system-level optimizations such as post-training quantization (PTQ/DRQ) are applied, reducing model size from approximately 9 MB to 2.5 MB and enabling faster inference with reduced memory usage. This hybrid edge cloud design balances accuracy, latency, and scalability, making it suitable for deployment in rural and resource constrained environments. Table 9 contains details on the technology used in cloud and edge computing based solutions and their purposes or applications.

6.5 Blockchain technology

Although blockchain technology is an idea originally developed for financial transactions, its applications have made inroads into agricultural challenges such as the detection of diseases in tomato plants. By its very nature that being decentralized makes the information at the farm level right up to the supply value chain accurate and transparent. Blockchain, combined with IoT sensors, drones, and satellite imagery can securely record

and distribute information regarding soil conditions, weather patterns, and symptoms of diseases. This, in turn, ensures real-time monitoring for early detection of the same diseases, which allows timely and resource efficient interventions (Marzougui et al., 2025). Blockchain further assures full traceability from farm to fork, making sure that only healthy tomatoes reach the supply chain, while it also contributes to the maintenance of consumer confidence in support of official compliance (Soy & Balkrishna, 2025). Blockchain technology enables farmers, researchers, and agricultural specialists to collaborate effortlessly by offering a secure platform for sharing and analysis.

Besides, smart contracts can automate various procedures, like notifications about the possibility of outbreaks when some criteria are satisfied, allowing for timely responses. This option contributes not only to diseases but also to optimization regarding resource use through a decrease in the level of pesticide application and taking resource saving technologies into account. Blockchain ensures data integrity and builds trust among stakeholders, and the use of blockchain is likely to greatly improve the front of disease management and detection within tomato farming (Vasireddy et al., 2024). Table 10 contains information on technology used in block chain, and their purposes or applications.

6.6 Comparative analysis of different advanced technologies for tomato plant cultivation

To identify the most suitable approach for real-world deployment, the methods discussed in Sections 6.1-6.5 are compared in Table 11 based on key functions, advantages and their challenges along with its predominant adoption scale.

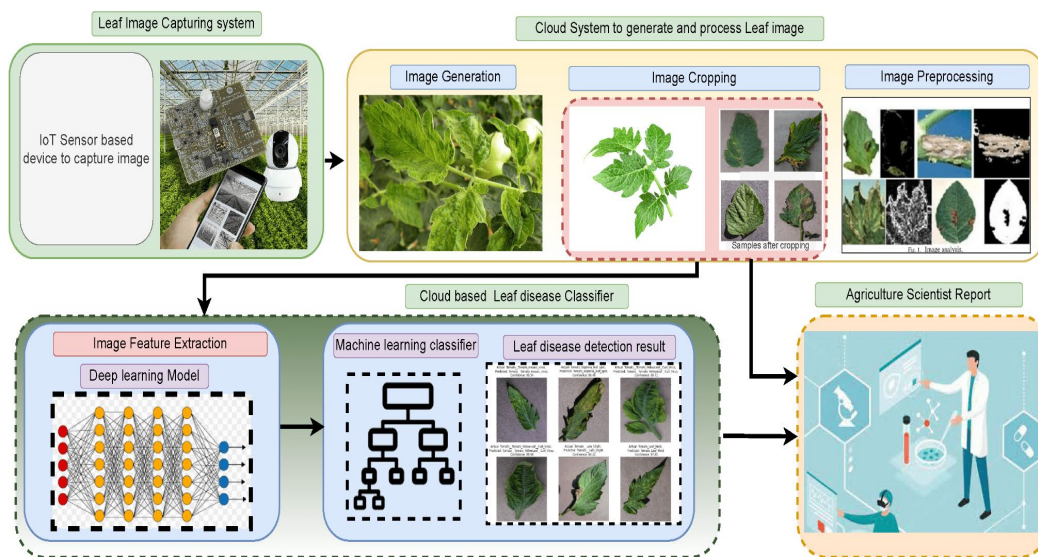


Figure 8. Cloud and IoT based detection system (Islam et al., 2022)

Table 9. Applications of Cloud and edge computing solutions for tomato farming

Ref	Technology used	Purpose/Application
(Gu et al., 2020)	Depthwise Separable Convolutional Neural Network (DSCNN)	Real-time monitoring and recognition of crop diseases using edge computing
(Suneja et al., 2022)	IoT, Machine Learning, Azure Custom Vision model, ThingSpeak	To monitor tomato plant health, including temperature, humidity, soil moisture, and sunlight, and detect and diagnose plant diseases.
(Rani et al., 2024)	Deep Learning Techniques, Transfer learning, Genetic Algorithm, Image Segmentation, Data Augmentation (DA)	Automated and accurate detection of tomato plant leaf diseases
(Zahid et al., 2024)	CNN (ResNet-18, EfficientNetB0, NasNetMobile, SqueezeNet, MobileNetV2, GoogleNet), Edge computing (Google Coral)	Tomato disease detection and classification using edge computing for real-time, low-power, and low-cost solutions.
(Zhang & Chen, 2023)	ANN, PNN, Logistic regression, GRNN, Edge computing	Detection of leaf curl disease in tomato crops using edge computing
(Sarvaiya et al., 2025)	MobileNetV2 with quantization (PTQ/DRQ) for crop disease classification on edge devices.	Model size reduced from ~9 MB to ~2.5 MB; accuracy ~92.23% vs ~94% original, shows trade-off and viability. Small sized models help in real world deployment
(Hayajneh et al., 2024)	TinyML framework for UAV + IoT sensor use in agriculture; transfer learning on edge devices.	Demonstrates edge computing + cloud/IoT combo for agriculture use case with reduced model size
(Nyakuri et al., 2025)	Lightweight CNN architecture (Tiny-LiteNet)	Achieved ~98.6% accuracy, inference time ~80 ms, designed for portable/edge usage

Table 10. Applications of block chain technology for tomato farming

Ref	Technology used	Purpose/Application
(Sudha et al., 2023)	Ethereum Blockchain, Solidity Programming Language	To manage issues in the tomato supply chain such as farmer loss, post harvest loss, and lack of traceability. The system provides clearness, traceability, direct sales, data integrity, smart contracts, and secure data storage.
(Grecuccio et al., 2020)	Blockchain, IoT, Smart Contracts	Food chain traceability, ensuring food safety and integrity, providing information to consumers about the origin and production of their food, reducing fraud and counterfeiting in the food industry
(Sangeetha et al., 2021)	IoT, LoRa, Arduino, Raspberry Pi, Azure SQL Database, Blockchain (Ethereum)	Classification, crop quality monitoring, and yield prediction. The system aims to eliminate intervention in the trade, thereby helping farmers get a fair price for their produce.

In the real world implementation, a hybrid which incorporates various advanced technologies is the best solution because each element has a specific role to play in management of tomato crop cultivation. The algorithms (ML and DL) employed as AI-based serve as the decision-making core, which are shown to be applicable to small, mid, and large farming systems when deployed on mobile, cloud, and platform platforms. Sensor and IoT systems are used as underlying infrastructure, which enables sustained monitoring of the field at all scales and facilitates decision making based on the data. Small-scale and large commercial farms are the most common ones using drones and UAVs, as they offer large-scale surveillance that is fast and justifies the operational costs of such investments warrant. Cloud and edge computing platforms are suitable mostly in mid and big-scale deployment with scalable data management and analytics. Large scale and export based farming are areas in which blockchain technology is relevant, since supply-chain transparency and traceability are of paramount importance. The given analysis based on the adoption scale offers a more deployment-oriented understanding of how the combination of the advanced technologies may be used in order to achieve sustainable and intelligent tomato production.

Table 11. Comparative analysis of advanced technologies for real world deployment

Technology	Key Functions in Hybrid system	Advantage	Challenge	Predominant Adoption Scale	Ref
AI (ML & DL)	Core decision-making layer Image-based disease identification and yield prediction	High accuracy, supports automation and predictive/recommendation maintenance	Requires large datasets and processing power	Small, Mid, Large (via mobile, cloud, and enterprise platforms)	(Hasan et al., 2019; Aversano et al., 2020; Islam, 2020; Medici et al., 2020; Ouhami et al., 2020; Luna-Benoso et al., 2021; Vallejo-Pérez et al., 2021; Vandana & Goel, 2021; Yoren & Suyanto, 2021; Bhatia et al., 2022; Islam et al., 2022; Tarek et al., 2022; Zhang et al., 2022; Zhou et al., 2022; Nyasulu et al., 2023; Alzahrani, 2025; Bala et al., 2025; Mezenner et al., 2025; Sharma et al., 2025; Sun et al., 2025)
Sensor & IoT Systems	Data acquisition & infrastructure layer Soil moisture, humidity, and temperature monitoring	Enables precision farming, real-time alerts, efficient resource use	Connectivity issues, requires frequent calibration	Small, Mid, Large farms	(Choi et al., 2013; Ampatzidis et al., 2017; Verma et al., 2018; Abdulridha et al., 2020; Hemming et al., 2020; Rezvani et al., 2020; Al-Mashhadani & Chandrasekaran, 2021; Darmawan et al., 2021; Thalluri et al., 2021; Chung et al., 2022)
Drones / UAVs	Monitoring & surveillance layer Aerial imaging, disease and nutrient monitoring, yield estimation	Real-time field monitoring, large coverage area, reduces manual labor	Weather-dependent, high initial and maintenance cost	Mid scale and Large / Commercial farms	(Thalluri et al., 2021; Kouadio et al., 2023)
Cloud & Edge Computing	Computational backbone Data collection, storage, and predictive analytics	Enables remote monitoring, scalable data access	Concerns over data privacy and latency	Mid scale and Large / Commercial farms	(Gu et al., 2020; Suneja et al., 2022; Zhang & Chen, 2023; Hayajneh et al., 2024; Rani et al., 2024; Zahid et al., 2024; Nyakuri et al., 2025; Sarvaiya et al., 2025)
Blockchain Technology	Post-harvest & supply-chain layer Supply chain traceability, data security, and transparency	Ensures trust and authenticity in product distribution	Complex setup, high operational cost	Large / Commercial and export oriented farms	(Grecuccio et al., 2020; Sangeetha et al., 2021; Sudha et al., 2023)

AI (Machine Learning and Deep Learning): The versatility of AI-based methods and scalability is the reason why these methods are implemented in small, mid, and large farming systems. Lightweight AI models embedded in mobile applications can be used to aid the diagnosis of diseases on a smartphone through the use of photos on a small-scale farm. Cloud-based AI platforms are beneficial to mid-scale farms, which abound in batch processing, predictive analytics, and advisory services. Enterprise-scale disease surveillance, predicting yields, and automating decisions are supported by enterprise-scale AI pipelines that connect to cloud infrastructure which are used in large and commercial farming systems. Such flexibility in computing environments justifies why AI is so widely used as the decision-making layer in managing tomato crops.

Sensor and IoT Systems: The use of sensor and IoT technologies is very universal in that they can be used in all sizes of farms as they can deal with basic agronomic needs like monitoring of soil moisture, temperatures, and humidity. Small farms are known to have low-cost sensors that are used in simple environmental monitoring and irrigation control whereas mid-scale farms combine several sensor nodes to produce zone-based precision farming. For large farm systems, the utilization of dense sensor networks and centralized data platforms can provide continuous and high-resolution monitoring. The modular and incremental nature of IoT deployments allow farmers to scale up their infrastructure based on the availability of resources and the size of their farm.

Drones and Unmanned Aerial Vehicles (UAVs): The main reason why drones and UAVs are only implemented in mid-scale and large farms is that they are expensive and require good weather conditions to function. In the case of small farms, drone technology is commonly restricted to shared solutions or service-based models, where third-party service organizations can conduct regular aerial surveys. Drones are used as cost-effective methods of identifying diseases, analyzing nutrient stress, and estimating yields at a larger scale in the mid-sized farms. When commodity-scale commercial farms embrace drones to eliminate reliance on labor and utilize drones to conduct field-wide surveillance at the opportune time, they become economically viable at scale.

Cloud and Edge Computing: These services are mostly used in medium and large-scale agricultural systems in which large volumes of data and real-time analytics are needed. Indirectly, small farms are gaining the benefits of cloud computing in advisory applications based on AI and centralized data storage services offered by third parties. Cloud platforms can be used by mid-size farms to remotely monitor, store data, and provide support to decision-making, and edge computing can be utilized to provide low-latency responses to time-sensitive processes like irrigation control. Hybrid cloud-edge architectures can serve to support large sensor networks, drone data processing, and enterprise-level analytics, which is why they should be adopted in large-scale deployments.

Blockchain Technology: The use of blockchain is mostly centralized in large-scale and export-oriented agricultural systems. It cannot be adopted in smaller farms because of the complexity of its infrastructure and the cost of its operations. Smaller farms are more likely to be involved in blockchain-based systems indirectly by cooperatives or aggregators, in which records on transactions and provenance information are centralized. Middle-range farms selectively use blockchain to assure quality as well as to supply chains as contracts. The implementation capability of blockchain, which includes traceability, data integrity, and farming listed food safety standards internationally, is most beneficial to large commercial farms and exporters as a strategic technology in post-harvest and supply-chain management.

A finite automata model that describes the life cycle of tomato plants and the associated disease states is shown in Figure 9. Starting with the seed stage, this model covers the germination, seedling, vegetative, flowering, and fruit stages. The plant's growth

trajectory is represented by the interconnected stages. The model incorporates possible disease states that can impact the plant at different phases of its lifecycle. The stages of seedling, vegetative, flowering, and fruit development can all be impacted by the disease state, which can impede normal progression. The model also incorporates a recovery state, which gives the plant the chance to manage disease and continue its growth.

This finite automata model is influential in understanding the dynamics of tomato plant growth and the critical points where diseases can manifest. By mapping out these stages and their interconnections, researchers and farmers can better identify and manage disease interventions. The detailed representation of disease states and recovery pathways bring out the importance of timely and effective disease management practices to ensure healthy plant development and optimal crop yields.

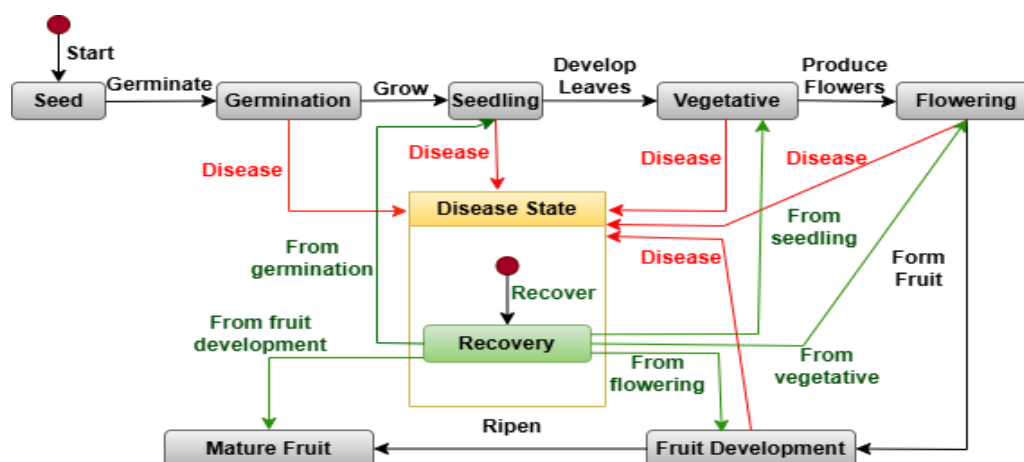


Figure 9. Finite automata model depicting the tomato plant life cycle and disease states

7. Conclusion and Future work

The paper was a comprehensive review of the latest developments in the field of tomato production and control of diseases, in particular, the consideration of the five research questions developed (RQ1-RQ5).

For RQ1, it was determined that there is a range of disease susceptibility in the tomato life cycle, with seedling-fruiting being the most sensitive phase since the infections in these stages have direct influence on yields and economic gains. This points out the need to have continuous monitoring as opposed to interventions that are stage isolated.

For RQ2, it revealed that regional and varietal variation in tomato farming especially in India posed a great variation in the morphology and the appearance of the plant. Therefore, heterogeneous datasets, Pusa Ruby and Vaishali, must be used to train disease detection systems to experience robustness across different varieties.

For RQ3, it was established that environmental conditions such as temperature (20-30° C), soil pH (approximately 7), humidity, and irrigation regimens have a significant impact on the occurrence of disease, which supports the significance of sensor-based monitoring systems to make risk assessment early.

For RQ4, it was found that the presence of fungal, bacterial, and viral diseases was one of the key limitations to tomato production. The weaknesses of conventional and

AI-based detection methods, especially concerning the capability to generalize across diverse field conditions and different stages of the life-cycle was also pointed out.

For RQ5, it could be concluded that more developed AI based methods, particularly deep learning models with transfer learning (e.g., VGG16, ResNet, and MobileNet variants) performed better on disease classification with reported accuracy between 92 and 99%. Such outcomes are mainly ascribed to the fact that convolutional neural networks are capable of learning discriminative visual features in different illumination, background noise and leaf orientations. Though the greatest effectiveness was observed in AI, the other complementary technologies, including drones and IoT systems, are essential in the process of real-time data collection. In addition, blockchain was shown to contribute to the supply chain in agriculture from the traceability and transparency perspective.

Overall, the findings indicate that integrated, hybrid smart farming frameworks combining AI-driven diagnosis with IoT-based sensing and drone-assisted surveillance offer the most practical and scalable pathway for sustainable tomato disease management. Future research should focus on translating high-accuracy models into deployable solutions through lightweight architectures, edge computing, and Generative AI-assisted decision support systems to ensure accessibility for farmers in real-world agricultural settings.

Furthermore, future research can be concentrated around existing limitations involving Generative AI (GenAI), and IoT-based environmental frameworks to build unified structures for proactive disease prevention, yield optimization and precision farm management. The limitation can be bridged in case of such innovation between technology and conventional farming, promoting collaborations between researchers, engineers, and farmers to fulfil the real potential of intelligent tomato farming.

8. Authors' Contributions

Gangadhar Shankarappa designed and conducted the research, analyzed data, and wrote the manuscript. Pramod Tumakuru Channabasavanna contributed to data interpretation, provided technical insights, and reviewed the manuscript for improvements. Both authors read and approved the final version of the manuscript.

9. Conflicts of Interest

The authors declare no conflicts of interest relevant to this research.

References

- Abdulridha, J., Ampatzidis, Y., Qureshi, J., & Roberts, P. (2020). Laboratory and UAV-based identification and classification of tomato yellow leaf curl, bacterial spot, and target spot diseases in tomato utilizing hyperspectral imaging and machine learning. *Remote Sensing*, 12(17), Article 2732. <https://doi.org/10.3390/rs12172732>
- Al-Mashhadani, Z., & Chandrasekaran, B. (2021). ROS-based robotic system for tomato disease and ripeness classification using convolutional neural networks. In *Proceedings of the IEEE 12th annual information technology, electronics and mobile communication conference (IEMCON)* (pp. 420-427). IEEE. <https://doi.org/10.1109/IEMCON53756.2021.9623183>
- Alizadeh-Moghaddam, G., Nasr-Esfahani, M., Nasr-Esfahani, A., & Mohammadbagheri, L. (2024). A suitable alternative to antifungal agents for the control of early blight disease

- *Alternaria alternata* of tomato. *Australasian Plant Pathology*, 53(2), 129-140. <https://doi.org/10.1007/s13313-024-00968-6>
- Alzahrani, M. (2025). Automated tomato defect detection using CNN feature fusion for enhanced classification. *Processes*, 13(1), Article 115. <https://doi.org/10.3390/pr13010115>
- Ampatzidis, Y., De Bellis, L., & Luvisi, A. (2017). iPathology: Robotic applications and management of plants and plant diseases. *Sustainability*, 9(6), Article 1010. <https://doi.org/10.3390/su9061010>
- Asia Farming. (2025). *Different types of agriculture subsidy in India: List of farming subsidy schemes by Government of India*. <https://www.asiafarming.com/different-types-of-agriculture-subsidy-in-india-list-of-farming-subsidy-schemes-by-government-of-india>
- Aversano, L., Bernardi, M. L., Cimitile, M., Iammarino, M., & Rondinella, S. (2020). Tomato diseases classification based on VGG and transfer learning. In *Proceedings of the IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor)* (pp. 129-133). IEEE. <https://doi.org/10.1109/MetroAgriFor50201.2020.9277626>
- Bala, M. N., Nayak, T. K., Palla, J. S. S. P., Nath, S. G., & Jeswanth, S. (2025). Plant leaf disease detection using MobileNetV2. *ITM Web of Conferences*, 79, Article 01021. <https://doi.org/10.1051/itmconf/20257901021>
- Bhatia, A., Chug, A., Singh, A. P., Singh, R. P., & Singh, D. (2022). A machine learning-based spray prediction model for tomato powdery mildew disease. *Indian Phytopathology*, 75, 225-230. <https://doi.org/10.1007/s42360-021-00430-3>
- Blancard, D. (2012). *Tomato diseases*. Academic Press.
- Chakravarthy, A. S., & Raman, S. (2020). Early blight identification in tomato leaves using deep learning. In *Proceedings of the international conference on contemporary computing and applications (IC3A)* (pp. 154-158). IEEE. <https://doi.org/10.1109/IC3A48958.2020.233288>
- Chavez-Martinez, O., Monjardin-Armenta, S. A., Rangel-Peraza, J. G., Mora-Felix, Z. D., & Sanhouse-García, A. J. (2025). Dataset of aerial photographs acquired with UAV using a multispectral (green, red and near-infrared) camera for cherry tomato (*Solanum lycopersicum* var. *cerasiforme*) monitoring. *Data in Brief*, 58, Article 111256. <https://doi.org/10.1016/j.dib.2024.111256>
- Chin, R., Catal, C., & Kassahun, A. (2023). Plant disease detection using drones in precision agriculture. *Precision Agriculture*, 24(5), 1663-1682.
- Choi, E. Y., Choi, K.-Y., & Lee, Y.-B. (2013). Non-drainage irrigation scheduling in coir substrate hydroponic system for tomato cultivation by a frequency domain reflectometry sensor. *European Journal of Horticultural Science*, 78(3), 132-143.
- Chou, H.-P., Huang, Y.-C., Lin, Y.-H., & Deng, W.-L. (2022). Selection, formulation, and field evaluation of *Bacillus amyloliquefaciens* PMB01 for its application to manage tomato bacterial wilt disease. *Agriculture*, 12(10), Article 1714. <https://doi.org/10.3390/agriculture12101714>
- Chung, I., Gupta, A., & Ogunfunmi, T. (2022). Remote crop disease detection using deep learning with IoT. In *Proceedings of the IEEE global humanitarian technology conference (GHTC)* (pp. 353-360). IEEE. <https://doi.org/10.1109/GHTC55712.2022.9910991>
- Cordon, G., Andrade, C., Barbara, L., & Romero, A.M. (2022). Early detection of tomato bacterial canker by reflectance indices. *Information Processing in Agriculture*, 9(2), 184-194. <https://doi.org/10.1016/j.inpa.2021.06.004>
- Darmawan, R. R., Rozin, F., Evani, C., Idris, I., & Sumardi, D. (2021). IoT and machine learning system for early/late blight disease severity level identification on tomato plants. In *Proceedings of the 13th International conference on information and communication technology and system (ICTS)* (pp. 288-293). IEEE. <https://doi.org/10.1109/ICTS52701.2021.9608788>

- Devaraj, H., Sohail, S., Ooi, M., Li, B., Hudson, N., Baughman, M., Chard, K., Chard, R., Casella, E., Foster, I., & Rana, O. (2024). RuralAI in tomato farming: Integrated sensor system, distributed computing and hierarchical federated learning for crop health monitoring. *IEEE Sensors Letters*, 8(5), Article 5501604. <https://doi.org/10.1109/LSSENS.2024.3384935>
- Devi, N. O., Devi, R. K. T., Debbarma, M., Hajong, M., & Thokchom, S. (2022). Effect of endophytic *Bacillus* and arbuscular mycorrhiza fungi (AMF) against *Fusarium* wilt of tomato caused by *Fusarium oxysporum* f. sp. *lycopersici*. *Egyptian Journal of Biological Pest Control*, 32, Article 1. <https://doi.org/10.1186/s41938-021-00499-y>
- Dhaliwal, M. S., Jindal, S. K., Sharma, A., & Prasanna, H. C. (2020). Tomato yellow leaf curl virus disease of tomato and its management through resistance breeding: A review. *Journal of Horticultural Science and Biotechnology*, 95(4), 425-444. <https://doi.org/10.1080/14620316.2019.1691060>
- Drishti IAS. (2023, July 1). *Tomatoes price volatility*. <https://www.drishtiias.com/daily-updates/daily-news-analysis/tomatoes-price-volatility>
- Egi, Y., Hajyzadeh, M., & Eyceyurt, E. (2022). Drone-computer communication based tomato generative organ counting model using YOLOv5 and DeepSORT. *Agriculture*, 12(9), Article 1290. <https://doi.org/10.3390/agriculture12091290>
- Elsharkawy, M. M., El-Okkiah, S., Elsadany, A. Y., Bedier, M. Y., Omara, R. I., Behiry, S. I., Hassan, S., & Abdelkhalek, A. (2022). Systemic resistance induction of tomato plants against tomato mosaic virus by microalgae. *Egyptian Journal of Biological Pest Control*, 32(1), Article 37. <https://doi.org/10.1186/s41938-022-00538-2>
- Gangadhar, S., & Sowmya. (2020). Real time environmental time series data analysis using Influx DB. *International Journal of Advanced Scientific Innovation*, 1(1), 1-5. <https://doi.org/10.5281/zenodo.4641703>
- Giri, V. P., Pandey, S., Srivastava, S., Shukla, P., Kumar, N., Kumari, M., Katiyar, R., Singh, S., & Mishra, A. (2023). Chitosan fabricated biogenic silver nanoparticles (Ch@BSNP) protectively modulate the defense mechanism of tomato during bacterial leaf spot (BLS) disease. *Plant Physiology and Biochemistry*, 197, Article 107637. <https://doi.org/10.1016/j.plaphy.2023.03.014>
- Girish, L., Gangadhar, S., Bharath, T. R., Balaji, K. S., & Abhishek, K. T. (2018). Crop yield and rainfall prediction in Tumakuru district using machine learning. *International Journal for Research in Engineering Application and Management*, 4(Special issue NCTFRD-2018), 61-65.
- Gleason, M. L., & Edmunds, B. A. (2005). *Tomato diseases and disorders*. Iowa State University Extension.
- Goap, A., Sharma, D., Shukla, A. K., & Krishna, C. R. (2018). An IoT based smart irrigation management system using machine learning and open source technologies. *Computers and Electronics in Agriculture*, 155, 41-49. <https://doi.org/10.1016/j.compag.2018.09.040>
- Grecuccio, J., Giusto, E., Fiori, F., & Rebaudengo, M. (2020). Combining blockchain and IoT: Food-chain traceability and beyond. *Energies*, 13(15), Article 3820. <https://doi.org/10.3390/en13153820>
- Gu, M., Li, K.-C., Li, Z., Han, Q., & Fan, W. (2020). Recognition of crop diseases based on depthwise separable convolution in edge computing. *Sensors*, 20(15), Article 4091. <https://doi.org/10.3390/s20154091>
- Gupta, S. K., Sharma, M., & Mukherjee, S. (2022). Buckeye rot of tomato in India: Present status, challenges, and future research perspectives. *Plant Disease*, 106(4), 1085-1095. <https://doi.org/10.1094/PDIS-04-21-0861-FE>
- Hasan, M., Tanawala, B., & Patel, K. J. (2019). Deep learning precision farming: Tomato leaf disease detection by transfer learning. In *Proceedings of the 2nd international*

- conference on advanced computing and software engineering (ICACSE 2019) (pp. 171-175). SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.3349597>
- Hassan, S. M., & Maji, A. K. (2022). Plant disease identification using a novel convolutional neural network. *IEEE Access*, 10, 5390-5401. <https://doi.org/10.1109/ACCESS.2022.3141371>
- Hayajneh, A. M., Aldalahmeh, S. A., Alasali, F., Al-Obeidollah, H., Zaidi, S. A., & McLernon, D. (2024). Tiny machine learning on the edge: A framework for transfer learning empowered unmanned aerial vehicle assisted smart farming. *IET Smart Cities*, 6(1), 10-26. <https://doi.org/10.1049/smc2.12072>
- Hemming, S., de Zwart, F., Elings, A., Petropoulou, A., & Righini, I. (2020). Cherry tomato production in intelligent greenhouses—sensors and AI for control of climate, irrigation, crop yield, and quality. *Sensors*, 20(22), Article 6430. <https://doi.org/10.3390/s20226430>
- Islam, M. T. (2020). Plant disease detection using CNN model image processing. *International Journal of Engineering Research and Technology*, 9(10), 291-297.
- Islam, M. S., Sultana, S., Farid, F. A., Islam, M. N., Rashid, M., Bari, B. S., Hashim, N., & Husen, M. N. (2022). Multimodal hybrid deep learning approach to detect tomato leaf disease using attention-based dilated convolution feature extractor with logistic regression classification. *Sensors*, 22, Article 6079. <https://doi.org/10.3390/s22166079>
- Kibriya, H., Rafique, R., Ahmad, W., & Adnan, S. M. (2021). Tomato leaf disease detection using convolution neural network. In *Proceedings of the 18th international Bhurban conference on applied sciences and technology (IBCAST)* (pp. 346-351). IEEE. <https://doi.org/10.1109/IBCAST51254.2021.9393311>
- Kouadio, L., El Jarroudi, M., Belabess, Z., Laasli, S.-E., Roni, M. Z. K., Amine, I. D. I., Mokhtari, N., Mokrini, F., Junk, J., & Lahlali, R. (2023). A review on UAV-based applications for plant disease detection and monitoring. *Remote Sensing*, 15, Article 4273. <https://doi.org/10.3390/rs15174273>
- Luna-Benoso, B., Martínez-Perales, J. C., Cortés-Galicia, J., Flores-Carapia, R., & Silva-García, V. M. (2021). Detection of diseases in tomato leaves by color analysis. *Electronics*, 10(9), Article 1055. <https://doi.org/10.3390/electronics10091055>
- Mani, S. D., Pandey, S., Govindan, M., Muthamilarasan, M., & Nagarathnam, R. (2021). Transcriptome dynamics underlying elicitor-induced defense responses against Septoria leaf spot disease of tomato (*Solanum lycopersicum* L.). *Physiology and Molecular Biology of Plants*, 27, 873-888. <https://doi.org/10.1007/s12298-021-00970-y>
- MarketResearch.biz. (2025, Jan 5). *Tomatoes market*. <https://marketresearch.biz/report/tomatoes-market/>
- Marzougui, F., Elleuch, M., & Kherallah, M. (2025). Blockchain and IoT in smart agriculture: Analysis, opportunities, challenges, and future research directions. *Journal of Information Assurance and Security*, 19(3), 104-119. <https://doi.org/10.2478/ias-2024-0008>
- Mazumdar, P., Singh, P., Kethiravan, D., Ramathani, I., & Ramakrishnan, N. (2021). Late blight in tomato: Insights into the pathogenesis of the aggressive pathogen *Phytophthora infestans* and future research priorities. *Planta*, 253, Article 119. <https://doi.org/10.1007/s00425-021-03636-x>
- Medici, M., Pedersen, S. M., Carli, G., & Tagliaventi, M. R. (2020). Environmental benefits of precision agriculture adoption. *Economia Agro-Alimentare*, 21(3), 637-656. <https://doi.org/10.3280/ECAG2019-003004>
- Mezenner, A., Lakehal, M. R., Arab, N., Nemmour, H., & Chibani, Y. (2025). Precision agriculture: Tomato disease classification via compact convolutional vision transformer. *Algerian Journal of Signals and Systems*, 10(2), 87-90. <https://doi.org/10.51485/ajss.v10i2.266>

- Ministry of Agriculture and Farmers Welfare, Government of India. (2024). *Second advance estimates of 2023-24 of area and production of horticultural crops*. <https://pib.gov.in/Pressreleaseshare.aspx?PRID=2022761>
- Narimani, M., Pourreza, A., Moghimi, A., Mesgaran, M., Farajpoor, P., & Jafarbiglu, H. (2025). *Drone-based multispectral imaging and deep learning for timely detection of branched broomrape in tomato farms*. <https://doi.org/10.48550/arXiv.2509.09972>
- National Horticulture Board. (2025a, Jan 20). *Tomato diseases*. <https://www.nhb.gov.in/pdf/vegetable/tomato/tom002.pdf>
- National Horticulture Board. (2025b, Feb 1). *Tomato: Post harvest management*. National Horticulture Board. <https://www.nhb.gov.in/pdf/vegetable/tomato/tom013.pdf>
- Nyakuri, J. P., Nkundineza, C., Gatera, O., Nkurikiyeyezu, K. & Mwitende, G. (2025). AI and IoT-powered edge device optimized for crop pest and disease detection. *Scientific Reports*, 15, Article 22905. <https://doi.org/10.1038/s41598-025-06452-5>
- Nyasulu, C., Diattara, A., Traore, A., Ba, C., Diedhiou, P. M., Sy, Y., Raki, H., & Peluffo-Ordóñez, D. H. (2023). A comparative study of machine learning-based classification of tomato fungal diseases: Application of GLCM texture features. *Heliyon*, 9(11), Article e21697. <https://doi.org/10.1016/j.heliyon.2023.e21697>
- Ouhami, M., Es-Saady, Y., Hajji, M. E., Hafiane, A., Canals, R., & Yassa, M. E. (2020). Deep transfer learning models for tomato disease detection. In A. El Moataz, D. Mammass, A. Mansouri, & F. Nouboud (Eds.). *Image and signal processing* (pp. 65-67). Springer. https://doi.org/10.1007/978-3-030-51935-3_7
- Pramanik, D., Shelake, R. M., Park, J., Kim, M. J., Hwang, I., Park, Y., & Kim, J.-Y. (2021). CRISPR/Cas9-mediated generation of pathogen-resistant tomato against *Tomato Yellow Leaf Curl Virus* and powdery mildew. *International Journal of Molecular Sciences*, 22(4), Article 1878. <https://doi.org/10.3390/ijms22041878>
- Press Information Bureau. (2024). *Final estimates of 2022-23 and first advance estimates of 2023-24 of area and production of horticultural crops*. <https://www.pib.gov.in/PressReleaselframePage.aspx?PRID=2012191>
- Qiao, K., Liu, Q., Huang, Y., Xia, Y., & Zhang, S. (2020). Management of bacterial spot of tomato caused by copper-resistant *Xanthomonas perforans* using a small molecule compound carvacrol. *Crop Protection*, 132, Article 105114. <https://doi.org/10.1016/j.cropro.2020.105114>
- Qi, S., Shen, Y., Wang, X., Zhang, S., Li, Y., Islam, M. M., Wang, J., Zhao, P., Zhan, X., Zhang, F., & Liang, Y. (2022). A new NLR gene for resistance to tomato spotted wilt virus in tomato (*Solanum lycopersicum*). *Theoretical and Applied Genetics*, 135(5), 1493-1509. <https://doi.org/10.1007/s00122-022-04049-4>
- Rahman, S. U., Alam, F., Ahmad, N., & Arshad, S. (2023). Image processing based system for the detection, identification and treatment of tomato leaf diseases. *Multimedia Tools and Applications*, 82, 9431-9445. <https://doi.org/10.1007/s11042-022-13715-0>
- Ramírez, V., Martínez, J., Bustillos-Cristales, M. D. R., Cataneda-Antonio, D., Munive, J. A., & Baez, A. (2022). *Bacillus cereus* MH778713 elicits tomato plant protection against *Fusarium oxysporum*. *Journal of Applied Microbiology*, 132(1), 470-482. <https://doi.org/10.1111/jam.15179>
- Rani, S. S., Kumar, C. H. M. S., Felicita, S. A., Ganesh, S. S., Choubey, A., & Anitha, R. (2024). Development and evaluation of a distinctive cloud-based artificial intelligence system using deep learning techniques (AISDLT) for accurate detection of tomato plant leaf diseases. *International Journal of Intelligent Systems and Applications in Engineering*, 12(12S), 538-552.
- Rezvani, S. M., Abyaneh, H. Z., Shamshiri, R. R., Balasundram, S. K., Dworak, V., Goodarzi, M., Sultan, M., & Mahns, B. (2020). IoT-based sensor data fusion for determining optimality

- degrees of microclimate parameters in commercial greenhouse production of tomato. *Sensors*, 20(22), Article 6474. <https://doi.org/10.3390/s20226474>
- Rudramurthy, V., Bharath, T. R., & Gangadhar, S. (2018). Applying cloud computing and wireless sensor technology for making smart agriculture. *International Journal for Research in Engineering Application and Management*, 4(special issue NCTFRD-2018), 38-42.
- Sagheer, A., Mohammed, M., Riad, K., & Alhajhoj, M. (2020). A cloud-based IoT platform for precision control of soilless greenhouse cultivation. *Sensors*, 21(1), Article 223. <https://doi.org/10.3390/s21010223>
- Sangeetha, M., Thejaswini, G., Shoba, A., Gaikwad, S. S., Amretasre, R. T., & Nivedita, S. (2021). Design and development of a crop quality monitoring and classification system using IoT and blockchain. *Journal of Physics: Conference Series*, 1964, Article 062011. <https://doi.org/10.1088/1742-6596/1964/6/062011>
- Sarvaiya, S., Dhole, P., & Nandwanshi, I. (2025). Crop disease classification for edge devices: A quantized MobileNetV2 approach. *International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering*, 13(10), 39-45. <https://doi.org/10.17148/IJIREICE.2025.131006>
- Saxena, R., Chauhan, S., Ghosh, D. K., Kumar, R., Kumar, D., & Das, A. (2023). Trends and trajectory of Indian agriculture. In P. K. Ghosh, A. Das, R. Saxena, K. Banerjee, G. Kar, & D. Vijay (Eds.). *Trajectory of 75 years of Indian agriculture after independence* (pp. 3-22). Springer. https://doi.org/10.1007/978-981-19-7997-2_1
- Shahi, T. B., Xu, C.-Y., Neupane, A., & Guo, W. (2023). Recent advances in crop disease detection using UAV and deep learning techniques. *Remote Sensing*, 15(9), Article 2450. <https://doi.org/10.3390/rs15092450>
- Sharma, J., Al-Huqail, A. A., Almogren, A., Doshi, H., Jayaprakash, B., Bharathi, B., Rehman, A. U., & Hussien, S. (2025). Deep learning based ensemble model for accurate tomato leaf disease classification by leveraging ResNet50 and MobileNetV2 architectures. *Scientific Reports*, 15, Article 13904. <https://doi.org/10.1038/s41598-025-98015-x>
- Sharma, K. K., Kumar, P., Kaushik, S., & Chaudhary, S. (2023). Enhancing soil health and productivity through conservation agriculture. *Vigyan Varta*, 4(11), 231-235.
- Sharma, P., Thakur, S., & Negi, R. (2019). Recent advances in breeding of tomato- A review. *International Journal of Current Microbiology and Applied Sciences*, 8(3), 1275-1283.
- Singh, D., Devappa, V., & Yadav, D. K. (2022a). Suppression of tomato bacterial wilt incited by *Ralstonia pseudosolanacearum* using polyketide antibiotic-producing *Bacillus* spp. isolated from rhizospheric soil. *Agriculture*, 12(12), Article 2009. <https://doi.org/10.3390/agriculture12122009>
- Singh, R., Ao, N. T., Kangjam, V., Rajesha, G., & Banik, S. (2022b). Plant growth promoting microbial consortia against late blight disease of tomato under natural epiphytotic conditions. *Indian Phytopathology*, 75, 527-539. <https://doi.org/10.1007/s42360-022-00464-1>
- Sotelo-Cardona, P., Lin, M.-Y., & Srinivasan, R. (2021). Growing tomato under protected cultivation conditions: Overall effects on productivity, nutritional yield, and pest incidences. *Crops*, 1(2), 97-110. <https://doi.org/10.3390/crops1020010>
- Soy, A., & Balkrishna, S. M. (2025). Blockchain integration in agriculture for transparent farm-to-fork supply chains: Leveraging IoT and decentralized identity for enhanced traceability and security. *SHS web of conferences*, 216, Article 01073. <https://doi.org/10.1051/shsconf/202521601073>

- Sudha, V., Aswini, D., & Akiladevi, R. (2023). Blockchain-based tomato supply chain management. In *Proceedings of the 2nd international conference on advancements in electrical, electronics, communication, computing and automation (ICAECA)* (pp. 1-5). IEEE. <https://doi.org/10.1109/ICAECA56562.2023.10201119>
- Sultana, F., Islam, M. R., Sabbir, M. A., & Hossain, M. M. (2020). Plant growth promotion and suppression of damping off in tomato by plant growth promoting *Rhizobacterium Bacillus amyloliquifaciens*. *Crops*, 5(1), 59-68. <https://doi.org/10.20448/803.5.1.59.68>
- Sun, H., Fu, R., Wang, X., Wu, Y., Al-Absi, M. A., Chen, Z., Chen, Q., & Sun, Y. (2025). Efficient deep learning-based tomato leaf disease detection through global and local feature fusion. *BMC Plant Biology*, 25, Article 311. <https://doi.org/10.1186/s12870-025-06247-w>
- Suneja, B., Negi, A., Kumar, N., & Bhardwaj, R. (2022). Cloud-based tomato plant growth and health monitoring system using IoT. In *Proceedings of the 3rd international conference on intelligent engineering and management (ICIEM)* (pp. 237-243). IEEE. <https://doi.org/10.1109/ICIEM54221.2022.9853170>
- Tarek, H., Aly, H., Eisa, S., & Abul-Soud, M. (2022). Optimized deep learning algorithms for tomato leaf disease detection with hardware deployment. *Electronics*, 11(1), Article 140. <https://doi.org/10.3390/electronics11010140>
- Thalluri, L. N., Adapa, S. D., Priyanka, D., N., Sri Nitaya, N., Yasmeen, Sarma, A. V. S. Y. N., & Venkat, S. N. (2021). Drone technology enabled leaf disease detection and analysis system for agriculture applications. In *Proceedings of the Second International Conference on Smart Electronics and Communication (ICOSEC)* (pp. 1079-1085). IEEE. <https://doi.org/10.1109/ICOSEC51865.2021.9591837>
- Tripathi, P., Kumar, N., Rai, M., & Khan, A. (2022a). Applications of deep learning in agriculture. In M. A. Khan, R. Khan, & P. Praveen (Eds.). *Artificial intelligence applications in agriculture and food quality improvement* (pp. 17-28). IGI Global. <https://doi.org/10.4018/978-1-6684-5141-0.ch002>
- Tripathi, R., Vishunavat, K., & Tewari, R. (2022b). Morphological and molecular characterization of *Clavibacter michiganensis* subsp. *michiganensis* causing bacterial canker in tomatoes. *Physiological and Molecular Plant Pathology*, 119, Article 101833. <https://doi.org/10.1016/j.pmpp.2022.101833>
- Vallejo-Pérez, M. R., Sosa-Herrera, J. A., Navarro-Contreras, H. R., Alvarez-Preciado, L. G., Rodriguez-Vazquez, A. G., & Lara-Avila, J. P. (2021). Raman spectroscopy and machine learning for early detection of bacterial canker of tomato: The asymptomatic disease condition. *Plants*, 10(8), Article 1542. <https://doi.org/10.3390/plants10081542>
- Vandana, & Goel, A. K. (2021). Plant leaves diseases recognition using convolutional neural network and five other hybrid models. In *Proceedings of the 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)* (pp. 1-9). IEEE. <https://doi.org/10.1109/ICCCNT51525.2021.9580025>
- Vasireddy, S. S. D. M., Yalagala, S., Sikha, J., Vullaganti, R., Repudi, R., Chandra, G. R., & Anand, D. (2024). Irrigation in precision agriculture using blockchain Ethereum based on IoT. In *Engineering Proceedings*, 66(1), Article 29. <https://doi.org/10.3390/engproc2024066029>
- Vasudeva, R. S., & Raj, J. S. (1948). A leaf-curl disease of tomato. *Phytopathology*, 38(5), 364-369.
- Verma, S., Chug, A., & Singh, A. (2020). Application of convolutional neural networks for evaluation of disease severity in tomato plant. *Journal of Discrete Mathematical Sciences and Cryptography*, 23(1), 273-282. <https://doi.org/10.1080/09720529.2020.1721890>
- Verma, S., Chug, A., & Singh, A. P. (2018). Prediction models for identification and diagnosis of tomato plant diseases. In *Proceedings of the international conference on advances in computing, communications and informatics (ICACCI)* (pp. 1557-1563). IEEE. <https://doi.org/10.1109/ICACCI.2018.8554842>

- Xia, J., Zhang, W., Zhang, W., Yang, Y., Hu, G., Ge, D., Liu, H., & Cao, H. (2021). A cloud computing-based approach using the visible near-infrared spectrum to classify greenhouse tomato plants under water stress. *Computers and Electronics in Agriculture*, 181, Article 105966. <https://doi.org/10.1016/j.compag.2020.105966>
- Yallappa, D., Ramasamy, K., Surendrakumar, A., Suthakar, B., Kumar, A. P. M., Kannan, B., & Muthusami, M. K. (2024). Improving agricultural spraying with multi-rotor drones: A technical study on operational parameter optimization. *Frontiers in Nutrition*, 11, Article 1487074. <https://doi.org/10.3389/fnut.2024.1487074>
- Yoren, A. I., & Suyanto, S. (2021). Tomato plant disease identification through leaf image using convolutional neural network. In *Proceedings of the 9th international conference on information and communication technology (ICoICT)* (pp. 320-325). IEEE. <https://doi.org/10.1109/ICoICT52021.2021.9527425>
- Zahid, A., Majeed, Y., & Ojo, M. O. (2024). *Standalone edge AI-based solution for tomato diseases detection*. <https://ssrn.com/abstract=4824801>
- Zhang, H., Zhang, L., Wu, H., Wang, D., Ma, X., Shao, Y., Jiang, M., & Chen, X. (2025). Unmanned-aerial-vehicle-based multispectral monitoring of nitrogen content in canopy leaves of processed tomatoes. *Agriculture*, 15(3), Article 309. <https://doi.org/10.3390/agriculture15030309>
- Zhang, N., Wu, H., Zhu, H., Deng, Y., & Han, X. (2022). Tomato disease classification and identification method based on multimodal fusion deep learning. *Agriculture*, 12(12), Article 2014. <https://doi.org/10.3390/agriculture12122014>
- Zhang, Y., & Chen, M. (2023). An IoT-enabled energy-efficient approach for the detection of leaf curl disease in tomato crops. *Wireless Networks*, 29, 321-329. <https://doi.org/10.1007/s11276-022-03071-0>
- Zhang, Z., Li, J., Zhang, Z., Liu, Y., & Wei, Y. (2021). Tomato endophytic bacteria composition and mechanism of suppressiveness of wilt disease (*Fusarium oxysporum*). *Frontiers in Microbiology*, 12, Article 731764. <https://doi.org/10.3389/fmicb.2021.731764>
- Zhong, X., Li, H., Cai, Y., Deng, Y., Xu, H., Tian, J., Liu, S., Hou, M., Weng, H., Wang, L., Ruan, M., Zhong, F., Zhu, C., & Xu, L. (2025). Early detection of tomato gray mold based on multispectral imaging and machine learning. *Horticulturae*, 11(9), Article 1073. <https://doi.org/10.3390/horticulturae11091073>
- Zhou, L., Song, C., Li, Z., & Kuipers, O. P. (2021). Antimicrobial activity screening of rhizosphere soil bacteria from tomato and genome-based analysis of their antimicrobial biosynthetic potential. *BMC Genomics*, 22(1), Article 29. <https://doi.org/10.1186/s12864-020-07346-8>
- Zhou, Q., Ma, L., Cao, L., & Yu, H. (2022). Identification of tomato leaf diseases based on improved lightweight convolutional neural networks MobileNetV3. *Smart Agriculture*, 4(1), 47-56. <https://doi.org/10.12133/j.smartag.SA202202003>